Deep Learning Frameworks for Multi-Task Prediction of Drug–Target Interactions and Adverse Effects

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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

This thesis presents a deep-learning framework for multi-task prediction of DTIs and ADRs in an integrated manner. The objective is to predict collectively if and how much a drug binds to a given protein target together with a confidence-ranked profile of side effects for the same drug. Joint prediction has been getting more attention since many side effects result from off-target or unintended interactions that cannot be observed by conventional single-task models. Contrary to conventional methods which decouple DTI and ADR prediction, obtaining evidence from a single data modality, our work focuses on multi-modal evidence: drug SMILES and 3D molecular graphs as well as protein sequences and AlphaFold-based 3D structures. Curated DTI and drug–ADR data sets are reconciled through a common drug identification number represented as RxNorm Id. So, we propose a joint adapter network with multi-head predictors used for DTI classification and multi-label ADR inference under the guidance of cross-modal alignment. The resulting system (a) focuses on calibrated, decision-relevant predictions (b) generalizes robustly to unseen drugs and proteins in the cold-start scenario, and (c) provides a reproducible cacheable pipeline from featurization to inference —with the goal of lowering screening costs, detecting early safety signals, and driving translational impact in computational drug discovery.

Keywords: *Drug–Target Interaction (DTI), Adverse Drug Reaction (ADR), Multi-Task Deep Learning, Multimodal Representation Learning, Contrastive Learning, Cross-Modal Alignment, ChemBERTa, ESM, Graph Neural Networks (GNNs), EGNN, GVP-GNN, TF-IDF Embeddings, Fusion Encoder, Adapter Networks, Computational Drug Discovery*

Dedication (Optional)

A dedication is the expression of friendly connection or thanks by the author towards another person. It can occupy one or multiple lines depending on its importance. You can remove this page if you want.

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First and foremost, we are thankful to the Almighty for giving us the chance and the path we needed to finish this project on time. Secondly, we would like to sincerely thank our thesis supervisor, Dr. Mohammad Golam Robiul Alam and our co-supervisor, Dr. Swakkhar Shatabda for their constant encouragement, mentorship, and guidance as we explored a challenging topic. Their relentless assistance and continuous input enabled us to face and overcome the challenges. Finally, we intend to use this opportunity to express our gratitude to every faculty member for their assistance and support during our stay at Brac University.

Table of Contents

List of Figures

List of Tables

Chapter 1

Introduction

This chapter presents the background of the study, the motivation and objectives of the research, and the reasons why a single framework to predict Drug-Target Interactions and Adverse Drug Reactions is necessary. Also, we will examine the theoretical context of the study, how it combined the various biological modalities of data, the overall scope of the study, and what issues it aimed to resolve.

1.1 Background

Drug-Target Interaction (DTI) prediction is an essential part of drug discovery in the modern era and the concept is to determine the interaction between the chemical compounds and the protein targets. The study of such interactions is of central significance since it defines the therapeutic potential of a drug in addition to the undesired biological effects of the drug. The classical methods of screening of the discovery of DTIs via the traditional laboratory techniques are accurate but time-intensive, costly and inefficient to conduct at the necessary scale to test millions of potential compounds. This has promoted the application of computational techniques that use machine learning and deep learning models on predicting the probability of interactions in a more effective way.

Defining whether a drug is bound to a protein is however not the only aspect of drug safety. A higher number of approved drugs, despite having good affinity to bind to their targets of interest, may result in Adverse Drug Reactions (ADRs) - unwanted or unintended physiological reactions including minor side effects or potentially fatal complications. ADR is a key challenge to pharmaceutical development and is a significant cause of drug post-marketing withdrawal. Computational prediction of such reactions has therefore become another significant field of study.

The recent advances in deep learning and specifically in multimodal structures able to incorporate heterogeneous biological information such as molecular graphs, SMILES representations, protein sequences and 3D structural data have provided new opportunities to improve both DTI and ADR prediction. However, current researchers tend to consider these two issues as two different tasks without the biological connection to each other. The mechanisms that regulate the drug-protein interactions are usually closely related to the mechanisms that lead to adverse effects. Therefore, as a combined predictor of DTI and ADR, it is possible that joint modelling of the two factors in a single framework could provide a more thorough perspective of the drug behaviour and potentially yield more interpretable and safer computational drug discovery systems.

This paper suggests a multi-task deep learning to forecast drug-target interactions and adverse drug reactions in a single architecture. Through cross-modal alignment and contrastive learning on a common representation space by multimodal data (chemical structure, protein sequence, and textual ADR data), we hope the model can accomplish both prediction tasks and be more trained to drugs and targets that were never seen previously.

1.2 Rational of the Study or Motivation

In the era of big data, drug discovery is becoming a data-rich activity that requires more than just the successful identification of therapeutic targets: it must increasingly account for the possibility of side effects that could be present at an earlier stage. Traditional computational pipelines address these issues separately. In Drug-Target Interaction (DTI) models, the goal is to predict molecular binding strength or affinity. In Adverse Drug Reaction (ADR) models, the aim is to find connections between drugs and side effects reported in clinical practice. While this division seems useful methodologically, it overlooks the biological continuity of a drug from its molecular action to its effects on human physiology.

The weakness of these approaches becomes obvious when considering safety issues after approval. Drugs with a good degree of target selectivity can still result in unexpected side effects as a consequence of interactions with other proteins or off-target binding. As discussed in [3] and [4] even more exact DTI predictors may leave downstream phenotypic consequences in the cold, if adverse outcomes are not accounted for at the prediction level. Likewise, chemistry-based, text co-occurrence based and ADR-based (ADR stands for Application Data Rate) predictors that predict only the occurrence of the side effect have limited interpretability and fail to explain why a particular ADR occurs.

The recent work by [5] and [26] suggests that structural and spatial context from proteins and compounds can highly assist the interaction prediction when exploited in conjunction with a graph-based or cross-attention design. But these studies have not yet been connected to molecular interactions that are related to the actual pharmacological effects in the real world, such as ADRs. The lessons learned from these experiments hint at the fact that multimodal embeddings can represent interaction, including the adverse effects.

This fact inspires the emergence of a DTI–ADR prediction model that can represent both dimensions of drug action, therapy, and side effects. A common model can use shared representations of molecular and biological characteristics to uncover interdependent patterns, which are otherwise obscured, when tasks are learned separately. In addition, the inclusion of contrastive learning [8] principles and enabling parameter sharing across chemical, protein, and phenotypic domains enables the model to learn from consistent mappings across modalities, leading to better generalization toward new compounds and targets.

Through integrating these two predictive fields, the present proposed approach seeks to advance a more comprehensive and interpretable computational paradigm for drug discovery, one capable of prioritising safer compounds, enabling faster repurposing of drugs, and decreasing expensive late-stage development failures.

1.3 Problem Statement

Despite the dramatic advances of deep learning for Drug–Target Interaction (DTI) prediction, prior computational modeling has been hindered by limitations in capturing intricate biological and chemical relationships that lead to both efficacy and side effects. Most previous works such as [3], [4] and [24] only predict the binding affinities or binary interaction labels, with no consideration of bridging these latent representations to the pharmacological responses. Their predictions, when accurate in vitro, typically do not generalize to reliable in vivo performances. Likewise, Adverse Drug Reaction (ADR) predictions based on text-mining or similarity-based approaches have not captured the molecular and target-level reasoning for how those reactions originated. This disassociation of DTI and ADR prediction results in a lack of knowledge regarding how the drug's binding profile relates to side effects, which is an essential gap when designing safer and more explainable drug discovery pipelines.

In order to bridge this gap, we propose to develop a unified framework that can simultaneously learn from DTI and ADR data, based on a shared representation that explores the relationships between drug binding patterns and the resultant adverse observations. This method faces three primary challenges: heterogeneity in biological data types, imbalance between positively and negatively labeled interactions, and lack of direct mapping between DTI and ADR signals provided by existing datasets. To this end, we aim to construct a multi-task deep learning model that synergizes multiple modalities within a shared representation guided by contrastive and cross-modal learning principles. This further allows interpretable and generalizable prediction of drug-target interactions and ADR potential.

1.4 Objective

The main goal of this research is to create a multi-task deep learning framework that can predict Drug-Target Interactions (DTIs) and Adverse Drug Reactions (ADRs) together using a single structure. The framework aims to bridge the current divide between therapeutic interaction modeling and adverse effect prediction by learning shared representations that reflect the underlying biochemical and structural dependencies among drugs, proteins, and clinical outcomes.

To achieve this goal, the study is guided by the following specific objectives:

- To construct an integrated dataset that aligns DTI and ADR data through standardized drug identifiers (e.g., RxNorm), ensuring consistent mapping between drug-protein pairs and associated adverse effects.

- To generate and compare multiple types of embeddings for both drugs and proteins using diverse encoders- ChemBERTa, EGNN and SMILES2Vec for chemical representations; ProtVec, ESM, GVP for protein representations, capturing both sequence-level and structural information.
- To design a modular shared adapter network that feeds into multi-head predictors, where one head performs DTI classification or binding affinity estimation, and the other performs multi-label ADR inference from the same shared latent space.
- To employ contrastive and cross-modal alignment learning to unify heterogeneous embedding spaces (chemical, structural, and textual) into a coherent shared representation that improves generalization to unseen drugs and protein targets.

Through these objectives, the study seeks to demonstrate that a multi-head, multimodal, and geometry-aware deep learning framework can effectively capture both therapeutic and toxicological dimensions of drug behavior—offering a more reliable and mechanistically grounded approach to computational drug discovery.

1.5 Methodology in Brief

This study is based on the systematic pipeline that incorporates the areas of dataset integration, generating multimodal embeddings, and training multi-task models. These steps involve curation and mapping of the DTI and ADR data sets, and the identifiers of the drugs are standardized via RxNorm mapping to offer some degree of uniformity across the data sets. Every drug protein pair in the DTI data is associated with the known adverse reactions and this is used to construct a single foundation in which they can be predicted together.

Feature representations are created using various biological modalities. SMILES2Vec and ChemBERTa are used to obtain drug embeddings, whereas ProtVec, ESM, GVP, and EGNN are used to obtain protein embeddings, which represent both sequence-level and 3D structure-based information. The encoding of ADRs is based on TF-IDF representations obtained with the help of text descriptions of adverse events.

These embeddings are then inputted to a common adapter network which learns to generate heterogeneous features in a common latent space. The model uses a multi-head architecture, in which one head predicts drug -target interactions (classification or affinity estimation), and the other predicts possible adverse reactions (multi-label regression). Contrastive learning and cross-modal alignment goals are used to increase the coherence of representation, with closely related drug-protein-ADR-triplets being embedded in the same space.

This model has been trained and tested on different drug and protein embedding combinations to determine the most successful one. The standard metrics of performance are used to measure performance, with special consideration on extrapolation of unobserved drugs and targets.

1.6 Scopes and Challenges

The suggested architecture has the potential of helping discover drugs through computational methods because it allows predicting Drug-Target Interactions (DTIs) and Adverse Drug Reactions (ADRs) at the same time, in one model. Its uses cover an extensive number of areas in pharmaceutical research such as drug repurposing, side-effect profiling and safety estimation of targets. With a mixture of multimodal sources of data, including chemical sequences, molecular graphs, and structure of a protein and even the textual description of ADR, the model will build a more comprehensive view of drug behavior that is expected to be generalizable to new targets and new molecules. Moreover, it is easy to extend it with other forms of embedding or even to new biological modalities since it is implemented in a modular fashion, and reconfigured to new biomedical data.

However, a lot of challenges are linked to this ambition. One of the common problems is data imbalance (the number of non-interaction drug-protein pairs is much larger than the positive interaction ones), which may result in biased learning for models. Another issue is the heterogeneity of data modality (chemical, structural and textual) that makes it hard to match embeddings in a common latent space. Moreover, since there is no one-to-one mapping between some DTIs and their ADRs, there is an indirect relational learning and careful loss design. Lastly, some of the high-dimensional 3D encoders like GVP, EGNN unavoidably result in weighty models which incur higher training costs or even need some sort of hardware acceleration.

Despite these limitations, the conceptual model proposed in this study offers a viable route to a truly multimodal and general-purpose predictive pharmacology system, contributing to safer and more expeditious drug discovery.

Chapter 2

Literature Review

This chapter initially covers the concepts, definition, and mechanisms of multiple machine learning (ML) and deep learning (DL) models to understand the baseline related to our paper. Secondly, several papers in relation to DTI and the ADR prediction were searched, including their methodology and pre-processing stages. Finally, the main results of those research papers are extracted: multiple frameworks on DTI and ADR technique preprocessing and prediction.

2.1 Preliminaries

2.1.1 Theories and Concepts

Drug–Target Interaction (DTI)

DTI is the biochemical action of a molecule (drug) binding to a protein target that results in a physiological or therapeutic effect. The ability to predict such interactions using computational approaches would be advantageous for the early identification of potential candidate drugs, without relying only on expensive lab results.

Binding Affinity and Specificity

How strongly a drug binds to its target, also known as binding affinity, is what determines the drug's potency; specificity represents how selectively it can bind to the intended target versus unintended targets. Both are key to understanding whether a drug works and what its side effects may be.

Ligand-Based and Structure-Based Paradigms

Two paradigms drove classical computational drug discovery. Ligand-oriented methods, like QSAR models, are based on the assumption that compounds sharing similar structural features possess similar biological properties. Structure-based techniques, such as molecular docking, model how a drug binds in three dimensions to the binding site of a target protein in order to predict interaction strength. Every such strategy relies a lot on handcrafted features and experimentally determined structures.

Data-Driven Learning in Drug Discovery

With the emergence of large bioactivity data sets, DTI prediction was recast as a supervised learning problem based on machine-learning techniques. Deep neural networks now directly extract novel, latent molecular representations from raw sequences or graphs, so that we no longer depend on expert-designed features and will have better generalization ability for unseen drug–protein pairs.

Adverse Drug Reactions (ADRs)

ADRs are unwanted and mostly adverse effects resulting from drug use. Most ADRs are due to drug off-target interactions that induce the drugs of interest on unintended proteins. This concept overlap between DTI and ADR prediction suggests a common biochemical basis of the therapeutic effect as well as of side effects.

Multi-Task Pharmacological Modeling

Especially since DTIs and ADRs are causally associated, joint modeling of DTIs and ADRs in an integrated framework enables a system to capture both binding associations as well as their clinical implications. In this way multi-task learning forms a theory-inspired bridge between molecular-level prediction and organism-level phenotypes.

2.1.2 Models and Methodologies

Conventional Computational Models

Earlier computational approaches in drug discovery were based on machine learning techniques such as Support Vector Machines (SVMs), Random Forests (RF), and k-Nearest Neighbors (k-NN). These models relied on predefined molecular descriptors like fingerprints or physicochemical features to classify drug–target pairs. While interpretable and computationally efficient, their performance was limited by the expressiveness of handcrafted features and their inability to generalize to unseen drugs or proteins.

Deep Learning-Based DTI Models

Deep learning introduced a major paradigm shift by enabling end-to-end feature learning from raw molecular and sequence data. Models such as [3] used convolutional neural networks (CNNs) to extract features from SMILES strings and amino acid sequences, while [6] and [24] expanded on this by integrating word-level and graph-based encodings. These architectures learn complex, nonlinear mappings between drug and protein representations, yielding superior binding-affinity prediction performance.

Graph Neural Networks (GNNs) for Structural Learning

To capture spatial and relational information inherent in molecular structures, Graph Neural Networks have become increasingly popular. Models like GVP-GNN and EGNN incorporate 3D geometric data by maintaining equivariance to spatial transformations, allowing them to model atomic-level interactions more accurately. These methods move beyond 1D sequence models, directly learning from the topology and geometry of molecular graphs derived from RDKit or AlphaFold structures.

Transformer Architectures and Pretrained Models

Transformer-based encoders such as ChemBERTa for molecules and ESM for proteins leverage self-attention to learn contextualized representations from large unlabeled corpora. By fine-tuning these pretrained embeddings for DTI prediction, researchers achieve improved generalization across diverse chemical and biological domains. Such models also serve as modular components that can be reused across multiple pharmacological tasks.

Multi-Task Deep Learning Frameworks

Modern research recognizes the interdependence between DTI and ADR prediction. Multi-task frameworks share a common backbone to learn generalized representations and employ separate output heads for distinct objectives—such as interaction classification and adverse effect regression. This structure enables knowledge transfer between related tasks, improving performance on both through shared inductive signals

Contrastive and Representation Alignment Methodss

Contrastive learning techniques are increasingly integrated to align heterogeneous data modalities. By minimizing the distance between embeddings of matching drug–protein pairs and maximizing it for non-matching ones, contrastive objectives encourage a coherent latent space. When extended to include ADR representations, this alignment helps the model learn how biochemical interactions translate into clinical outcomes.

Proposed Multi-Head Architecture

Building upon these methodologies, the proposed framework in this thesis introduces a shared adapter network that integrates multimodal embeddings—ChemBERTa and SMILES2Vec for drugs, ESM and GVP-GNN for proteins, and TF-IDF for ADRs. The network produces unified feature vectors that feed into two specialized heads: one for DTI prediction (classification or affinity regression) and another for ADR ranking (Huber/MSE-based regression). The model jointly optimizes DTI, ADR, and contrastive losses, enabling cross-modal learning between chemical, biological, and clinical domains

2.1.3 Embedding and Data Representation Techniques

Molecular Embeddings for Drugs

Drug molecules are typically encoded by their Simplified Molecular Input Line Entry System (SMILES) strings, which capture the atomic composition and interaction in a linear, text-based abstract. These sequences are then passed into deep learning models such as SMILES2Vec and ChemBERTa [7] etc., to map them to high-dimensional representations, which are able to capture structural and chemical semantics. SMILES2Vec gains distributed representation by introducing infrastructure based on the RNN or CNN models, and ChemBERTa exploits context information between atoms and substructures by leveraging the Transformers. These embeddings allow models to learn useful representations of molecular similarity, polarity, and reactivity based on pattern matching instead of handcrafted descriptors.

3D Structural Representations of Drugs

In addition to 1D sequence encoding, 3D coordinates of molecular structures (from conformers generated by RDKit) contain more complete spatial information. Graph-based encoders such as EGNN (Equivariant Graph Neural Network) directly act on such 3D graphs, learning features invariant under rotation and translation. Including such geometric embeddings enables the model to encode molecular shape, charge distribution, and interaction potential, which is essential for accurate binding affinity prediction.

Protein Sequence Embeddings

Protein targets are primarily represented through their amino acid sequences. ProtVec and ESM (Evolutionary Scale Modeling) are widely adopted sequence encoders. ProtVec uses skip-gram models inspired by natural language processing to learn representations of overlapping k-mers, capturing evolutionary motifs and functional similarities. ESM, a large-scale Transformer-based model trained on millions of protein sequences, produces contextualized embeddings that encode secondary structure and residue-level interactions, providing a robust basis for downstream DTI prediction. [1]

Protein Structural Embeddings

With the advent of AlphaFold, 3D structural data for proteins has become widely accessible. Graph-based models like GVP-GNN (Geometric Vector Perceptron GNN) effectively encode these structures by processing both scalar (amino acid features) and vector (spatial orientation) information. This dual representation allows the model to understand geometric dependencies within protein structures, enabling it to capture the spatial complementarity between drug molecules and their binding sites.

Adverse Drug Reaction (ADR) Textual Embeddings

ADR information typically comes from clinical narrative, drug labels, or pharmacovigilance databases. There are some textual vectorization techniques for representing ADRs in a machine-readable manner, such as the TF-IDF (Term Frequency–Inverse Document Frequency). TF-IDF measures for the significance of each adverse effect term in the context of its frequency among drugs, which can be interpreted as numerical embedding that is appropriate for multi-label regression. These representations afford the model the ability to connect molecular characteristics with observed clinical outcomes.

Unified Representation Using Common Adapter Networks

A shared projection technique is essential to obtain a unified representation of heterogeneous embeddings from drugs, proteins, and ADRs. We then share an adapter network that projects all input modalities into the same 512-dimensional latent space. This allows us to have a dimensionally compatible semantic correspondence between embeddings before they interact with task-specific heads. The shared space acts as a basis for contrastive alignment and multi-task prediction, allowing the model to learn cross-domain correlations (i.e., on how structural changes in a molecule relate to binding likelihood and adverse reactions).

2.2 Review of Existing Research

Ye et al. (2021)[16] addressed that DTI prediction is one of the key problems for drug discovery. A major challenge in DTI prediction is that both drugs and targets have complex graph based structures and, hence, are not amenable to traditional deep learning methods such as deep neural networks (DNNs) and convolutional neural networks (CNNs). The authors make a proposal of a multiple output graph convolutional network (MOGCN) which has the ability to enhance learning capabilities and increase the accuracy of DTI prediction. The method introduces auxiliary classifier layers with effective gradient propagation and effectively optimizes feature use at multiple levels. Moreover, the study uses a new graph calculation method which takes advantage of label information and manifold learning to form a meaningful adjacency matrix. The approach involves using stacked autoencoders (SAE) to obtain low level features, concatenating features and using graph convolutions. The proposed MOGCN was evaluated on five benchmark datasets, NR, GPCR, IC, E, and DB, and compared with baseline DNN baseline and original GCN. Experimental results show that MOGCN performs better than existing methods, and especially so when SAE based low level feature extraction is used. Based on the obtained results, MOGCN can tackle the gradient vanishing problem and at the same time improve the graph structures for DTI prediction, resulting in improved classification performance. The authors also discuss future work in increasing the depth and width of GCNs using residual network techniques and improving the graph calculation framework to capture more accurately relationships between interacting and non-interacting drug-target pairs.

Thafar et al. (2020)[9] addressed the challenge of silico drug target interaction (DTI) prediction as it is crucial for drug discovery and repurposing. Currently, the known computational methods suffer from an inordinate amount of false positive rates, and in short supply are experimentally validated DTI pairs. In order to solve this, the authors proposed DTiGEMS+, a framework that combines Graph Embedding, Graph Mining, and Similarity-based techniques to predict DTIs as a link prediction problem in a heterogeneous network. In this regard, the proposed approach builds a larger network with available DTIs, the drug–drug similarity network, and the target–target similarity network, which reflects a wider view of the drug target relationships. Multiple computational strategies have been employed in DTiGEMS+, namely similarity fusion, graph embedding using node2vec, path based feature extraction and machine learning classifiers like neural networks, random forests and

AdaBoost for final classification. DTiGEMS+ was evaluated on four benchmark datasets (NR, GPCR, IC, and E) and proved the highest average AUPR (0.92), 33.3% relative error rate reduction over the state of the arts. Biomedical literature and databases such as DrugBank and KEGG were used by the authors to validate novel predicted DTIs. As a future work, they propose improving the model by adding more similarity measures in the model, fine tuning the negative sampling strategies and enabling the framework to predict interactions for brands and new drugs and targets.

Mahmud et al. (2019)[5] addressed the challenge of drug discovery as a costly and time consuming process that is grounded on wet-lab experiments. A major limitation of DTI datasets and thus classification accuracy is the class imbalance issue identified in the study. In order to solve this issue, they came up with a high throughput computational model iDTi-CSsmoteB, which uses drug chemical structures and protein sequences to enhance DTI identification. Protein sequence features are extracted from PSSM-Bigram, AM-PseAAC, and dipeptide PseAAC descriptors and drug chemical structures are represented using molecular substructure fingerprint (MSF). The synthetic minority over-sampling technique (SMOTE) was utilized to handle the class imbalance problem and the XGBoost was used as the main classifier to predict DTIs. It was evaluated on four benchmark datasets (enzyme, ion channel, GPCR, and nuclear receptor) by means of five fold cross validation. The experimental results showed that iDTi-CSsmoteB achieves better area under the ROC curve (auROC) than the other existing methods, which verified the effectiveness of the proposed feature extraction techniques, data balancing methods, and classification model. Further, the study proposes that the model can be used in drug repurposing and novel drug discovery. As future works, semi supervised learning techniques and introducing precision recall metrics can be considered to improve the performance further.

Zhan et al. (2020)[10] study aimed to overcome the challenge of predicting drug–target interactions (DTIs) due to their importance in drug discovery and understanding biological processes. Identifying DTIs is costly, time consuming and labor intensive, and reliable computational methods are required to fill the gap. An ensemble learning approach using Local Optimal Oriented Pattern (LOOP), Position Specific Scoring Matrix (PSSM), and Rotation Forest (RF) is proposed by the authors to predict DTIs effectively. In particular, protein sequences are converted by PSSM while preserving evolutionary information, feature vectors are then extracted by LOOP from PSSM to reflect local patterns, and drug molecules are represented by fingerprint features. Finally, RF is applied as the classification model to infer DTIs. Four benchmark datasets, enzyme, ion channel, G protein-coupled receptors (GPCRs), and nuclear receptor, are used to evaluate the model with high average prediction accuracies of 89.09%, 87.53%, 82.05%, and 73.33% respectively. A comparison is performed with the state of the art methods like Support Vector Machine (SVM), K-Nearest Neighbor (KNN), and it is shown that the proposed approach is superior. Additionally, the study shows that the method enhances predictive performance but it relies on manual feature extraction that brings noise as well as ignores global structural information. To further boost the prediction accuracy of DTI, the authors propose that more robust feature extraction and augmentation or boosting of classification models would help.

Thafar et al.(2022)[19]addressed that drug-target binding affinity (DTBA) prediction is crucial for drug repositioning and virtual drug screening.. Most existing methods are based on three dimensional (3D) structural data of targets, which are frequently not available, or difficult to obtain. The authors propose Affinity2Vec, a regression based model that formulates the DTBA problem as a graph based task without the need for 3D structural information to overcome the above limitation. The method builds a weighted heterogeneous graph using drug-drug similarity, target-target similarity, and drug-target binding affinities. It is based on a mix of feature representation learning, graph mining, machine learning techniques. More specifically, drug representations are obtained from Simplified Molecular-Input Line-Entry System (SMILES) embeddings and protein sequences are encoded as ProtVec embeddings. Graph based meta path scoring and embedding based feature concatenation are applied in feature extraction, and the final prediction is made by Extreme Gradient Boosting (XGBoost). The model is evaluated against benchmark datasets Davis, KIBA and the PDBBind Refined, and performs better than existing state-of-the-art methods. It is evaluated using mean squared error (MSE), concordance index (CI) and area under the precision recall curve (AUPR) and found to be effective. Although Affinity2Vec has good predictive power, the authors concede that it does not have the ability to predict interactions of new drugs or targets and that it could be improved off future work with more large scale datasets and other molecular representations.

Yang et al. (2024)[29] proposed MINDG (Multi view Integrated Learning Network) which is a Multi view Integrated Learning Network with deep learning and graph learning to improve the drug target interaction (DTI) prediction. Traditional approaches suffer from the use of first order neighboring nodes in graph based models and lack of combination of both sequence and structural features of drugs and targets. To solve these problems, we propose a three steps framework, first, we use a hybrid deep network to extract the sequence based feature of the drugs and the targets; second, we use a high order graph attention convolutional network (HOAGCN) to discover deeper structural relationships. And, lastly, we use a multi view adaptive integrated decision module (MAIDM) to refine the prediction by combining the outputs from the first two steps. The evaluation of the model was done on the BindingDB dataset and DAVIS dataset for which it showed better performance than the state of art methods DeepCDA, TripletMultiDTI, graph attention networks (GAT) in terms AUPRC, AUROC, F1 Score. Experimental results demonstrate that MINDG is effective in combining deep learning and graph learning to well capture both intrinsic properties of molecules and relational interaction from outside. However, the authors concede that MINDG does not exploit all available intrinsic structural information of drugs and proteins and propose to further improve it in future work using more advanced graph learning techniques. Furthermore, the absence of wet-lab validation is also a limitation and future studies will involve the incorporation of experimental verification of the model to increase its reliability.

Ren et al.(2023) [23] developed DeepMPF which is a deep learning framework for DTIs prediction by using a multi-modal representation with meta-path semantic analysis. However, traditional DTI prediction methods are difficult to work with as it requires tremendous biological experiments and single modal computational models are limited to capture complex association behaviors in heterogeneous biological networks. In order to address these issues, DeepMPF integrates protein–drug–disease heterogeneous networks, and extracts feature information from three modalities: sequence, heterogeneous structure, and similarity. Six high order semantic representative meta-path schemas are designed to preserve hidden structural information by describing high order relationships. For the drug sequences and 3-mer sparse matrices used for the proteins, sequence embeddings are gained via natural language processing (NLP) tools; for the structural and the similarity features, Smith-Waterman scores and SIMCOMP similarity are used for this purpose. In order to generate comprehensive feature descriptors, the framework applies joint learning to improve predictive accuracy. DeepMPF is evaluated against state-of-the-art methods on four benchmark datasets, and it achieves better results than those methods in drug repositioning experiments on both COVID-19 and HIV. Moreover, the predictions of it are validated by molecular docking experiments, whose reliability is hereby reinforced. The authors admit that some biological features are not captured and recommend generalization through the addition of self attention mechanisms and larger datasets. Future work will focus on scaling up DeepMPF's effectiveness particularly in drug discovery applications.

Wang et al. (2022) [20] proposed the RoFDT model to overcome the challenge in predicting drug–target interactions (DTIs) that are critical in drug discovery and development. Because finding DTIs with traditional wet-lab methods is laborious and costly, computational models are used to predict large amounts of potential DTIs. The authors created RoFDT, an improvement of DTI prediction accuracy with the integration of feature weighted Rotation Forest (FwRF) with protein sequence and drug molecular structure information. The protein sequences were numerically encoded by Position Specific Score Matrix (PSSM) and the features were extracted by using Pseudo Position Specific Score Matrix (PsePSSM). Drug molecular structures were represented by molecular fingerprint descriptors in the meantime. The FwRF classifier was fed these features for prediction. The model was tested on four benchmark datasets, namely Enzyme, GPCR, Ion Channel and Nuclear Receptor, and the accuracy scores were 91.68%, 88.11%, 84.72%, 78.33% respectively. The result of RoFDT was found superior in its performance compared to support vector machines (SVM) and other existing models with 7 out of the top 10 predicted DTIs validated by the SuperTarget database. However, the study did stipulate some limitations of its effectiveness, like the dependence on PSSM for protein sequence representation and the manual feature selection. Eventually, improved automation will be provided; feature extraction techniques will be refined; and the model will be made more sensitive to newly found drug targets.

Mesrabadi et al. (2023)[21] where a computational model for DTI prediction was developed to overcome the high cost, time and complexity of experimental methods. The three main phases of their approach are feature extraction, feature selection and classification. The authors extracted various protein sequence features like EAAC, PSSM, etc., and also fingerprint features from drugs to form a complete feature representation in the feature extraction phase. However, the extracted data is high dimensional, a wrapper feature selection method called IWSSR was utilized in the next phase to eliminate redundant and less informative features. The Rotation Forest model was used to enhance predictive performance by classifying the selected features. This work innovates by combining different feature extraction techniques with IWSSR based feature selection and finally optimized classification. The model was evaluated on the gold standard datasets (enzyme, ion channel, G protein coupled receptor, and nuclear receptor) with accuracy of 98.12%, 98.07%, 96.82%, and 95.64% respectively. The high predictive performance of these results shows that the model is compatible with other state of the art methods. A key point of the study is based on the effectiveness of computational methods for DTI prediction, without explicitly mentioning limitations or future research directions.

Wu et al. (2021)[14] developed BridgeDPI, a novel graph neural network (GNN) based methodology to improve the prediction of drug–protein interaction (DPI) and to address the issues of existing computational methods such as poor quality of protein-protein associations (PPAs) and drug-drug associations (DDAs). The proposed model utilizes hyper nodes as bridges to connect proteins and drugs to enhance representation learning of PPAs and DDAs in an end-to-end supervised manner. Drug representations are created using molecular fingerprints and sequence-based features, while protein sequences are processed using k-mer features and convolutional neural networks (CNNs). Subsequently, these embeddings are fed into a GNN to combine hyper nodes for learning the interactions of drugs and proteins more accurately. Evaluation of BridgeDPI on various datasets such as customized BindingDB, C.elegans, and human datasets showed better performance than state-of-the-art models with AUC values of 0.989 for seen proteins and 0.952 for unseen proteins in BindingDB, 0.995 in C.elegans dataset, and 0.990 in human dataset. Moreover, BridgeDPI was also validated in a real world case study, by predicting interactions between viral proteins of COVID-19 and antiviral drugs, and the results matched those of the World Health Organization and the FDA, which further validated the practical usefulness of BridgeDPI. It is acknowledged by the authors that benchmark datasets suffer from data bias, resulting in high artificial performance metrics, and they conduct cross validation experiments on independent datasets to avoid this disadvantage. Results indicate that BridgeDPI can improve DPI prediction by using hyper-nodes, and thus may aid in drug discovery.

Öztürk et al. (2018)[3] developed DeepDTA, a deep learning based model for the drug-target binding affinity prediction as the traditional binary classification methods have the limitations of not predicting the strength of the drug-target interactions. Different from existing methods that depend on 3D protein-ligand complex structure or 2D molecular representation, DeepDTA uses only drug and protein sequence information and models them as CNN to predict binding affinity values. The model uses two separate CNN blocks to represent the drug SMILES and protein sequences and extracts high level feature representations before feeding them into the fully connected layers for affinity prediction. As a comparison test, the performance of DeepDTA was evaluated on the Davis and KIBA benchmark datasets and it already surpassed the traditional methods including KronRLS and SimBoost. DeepDTA performed better on the KIBA dataset, with a Concordance Index (CI) of 0.863, higher than that of KronRLS (CI: 0.782) and SimBoost (CI: 0.836) with statistical significance (P-value ¡ 0.0001). Results indicate that the use of CNNs to model sequence information helps in capturing interaction patterns which helps in improving the prediction of binding affinity. The authors recognize that CNN architectures may not be the best way to model the sequential dependencies in protein sequences, and therefore suggest that Long Short-Term Memory (LSTM) networks would improve predictive accuracy. The future work will also include integration of ligand based protein representations and investigating the alternative deep learning architectures to improve more drug target interaction prediction.

An et al., (2021)[11] developed a computational method named WELM SURF that increases drug target interaction (DTI) prediction accuracy by considering protein evolutionary information and molecular structure of the drugs. The computational approaches described in the study aim to address the limitations of experimental DTI validation, which can be very costly and time-consuming. To extract the evolutionary features from protein sequences, WELM-SURF uses Position Specific Scoring Matrix (PSSM), and to further refine the key sequence attributes from PSSM, SURF is used. Molecular substructure fingerprints are utilized as a means of creating feature vectors for drug representation. The Weighted Extreme Learning Machine (WELM) is used to classify DTIs because of its capability of efficiently optimizing the loss function while keeping the fast training speed and strong generalization performance. Evaluation of the model on four benchmark datasets, enzymes, ion channels, G-protein-coupled receptors (GPCRs), nuclear receptors was performed using fivefold cross validation. The accuracy scores for WELM-SURF are 93.54%, 90.58%, 85.43% and 77.45%, which is better than the well known traditional classifiers such as extreme learning machine (ELM) and support vector machine (SVM). It is demonstrated via comparative analysis with existing methods that WELM-SURF has much better feature extraction and prediction ability for DTIs. The authors stress that WELM-SURF is robust and efficient for large scale bioinformatics applications. Additionally, they point towards future work in certain areas, including improvements in feature extraction methodologies and the use of advanced machine learning algorithms which may improve predictive accuracy.

Liu and Paliwal (2024) [26] present DualBind: a deep learning framework for protein–ligand binding affinity prediction where they tackle the problems of prior models that only utilize labeled data or assume a Boltzmann-distributed dataset that might not be viable in reality. However, traditional supervised models utilize mean squared error (MSE) loss and tend to overfit in the case of lack of the labeled data. On the contrary, unsupervised methods such as denoising score matching (DSM) are based on generative modeling and unable to predict the absolute affinity values. DualBind first does a dual-loss integration combining MSE loss for the prediction of accurate absolute affinity and DSM loss to enhance the generalizability by learning from both labeled and unlabeled data, and then integrates two approaches in a dual-loss framework. The model was trained on the PDBbind v2020 refined dataset and evaluated on the CASF-2016 benchmark, achieving a Pearson correlation (Rp) of 0.757, an RMSE of 1.461, a Spearman correlation of 0.742 and performs better than AutoDock Vina, SimBoost and DSM-only models. Experiments also showed that DualBind is capable of leveraging the unlabeled data, sustaining high predictive performance with only half of the labeled dataset and surpassing purely supervised models. The paper by the authors emphasizes the ease of working with both labeled and unlabeled data on DualBind, and suggests future directions for research to be on hybrid training strategies for high performance models as well as their research on pretraining to improve model performance.

In order to predict DTIs, Zhao et al. (2024)[31] proposed a framework called MSI DTI, to integrate multi source information, i.e., sequence based biological features and network based features from bioinformatics knowledge graphs. The model has a drug representation module that uses 5 different types of representation (MACCS keys, Morgan fingerprints, Avalon fingerprints, Mol2Vec, and Graph2Vec) which are then combined to form a unified vector. This contains information like chemical substructure, connectivity patterns in molecular topology and molecular representation. After that the model builds a Drug-Target Knowledge Graph (DTKG) and uses multi head self attention with residual connection to capture both low and high order interaction information. Moreover, they used a pre-trained language model BioBERT to extract entities and relationships. The model also has a protein representation layer which employs two graph based embedding approaches for feature extraction from the heterogeneous biological networks. One of them is deep learning with random walks where they use the AttentionWalk algorithm to obtain node information. The other one is deep learning without random walk where they used combination-based multi-relation graph convolutional networks(CompGCN) which gave further insight into the representation and relationship of the edges in the entity representation when analyzing the node representation. After that the combined features are sent to the feature interaction and output module where they use PCA to decrease the dimension of the feature vector while retaining as much information as possible. By using a multi-head self-attention mechanism the model detects intricate relations between features while separate attention heads redirect their attention to different feature subspaces. The model combines all head outputs together with a mechanism that maintains primary features by using residual connections.

In the research conducted by Yan et al. (2021)[15] introduced CNNEMS, a deep learning model designed to predict drug-target interactions (DTIs) by integrating protein evolutionary data and molecular structural information. Identifying DTIs is important in the drug discovery process but biological experiments for this purpose are time consuming, costly and often with high false negative rates. In order to overcome these challenges, CNNEMS employs features extracted by a Convolutional Neural Network (CNN) and a classification using an Extreme Learning Machine (ELM). First, conversion of target protein sequences into Position-Specific Scoring Matrices (PSSM) and visualization using molecular fingerprint for drug compound representation are performed to capture biological evolutionary features. The CNN layers are then applied on the extracted feature vectors to extract abstract hidden features, which are finally classified using ELM algorithm. To validate the model, prediction accuracies of 94.19%, 90.%, 87.95%, and 86.11% was tested on enzymes, ion channels, GPCR, and nuclear receptor datasets, respectively. The higher accuracy and AUC scores in comparison to those of existing DTI prediction methods confirmed CNNEMS's superior performance. The authors note that although CNNEMS performs well, increased performance of predictive performance can be obtained with improved deep learning algorithms. Future work will refine the model by optimizing the methods of feature extraction and including more types of biological data in order to increase the robustness of the DTI predictions.

The research by Li et al. (2019)[4] presented deepatom as their data-driven system that applies a 3D-CNN structure to predict protein-ligand binding affinity. The DeepAtom framework extracts patterns of interaction by processing voxelized data from protein-ligand structures as opposed to physics-based and empirical and knowledge-based scoring functions which need substantial biological feature engineering. The system decreases bias linked to manually made descriptors through effective prediction models that produce generalized outcomes. The DeepAtom procedure includes three sequential stages starting from voxelization that applies feature channel encoding to a 3D grid box positioned around the ligand atom. The framework employs a light-weight 3D-CNN design that obtains interaction characteristics by utilizing depthwise separable convolutions with an efficient structure to reduce overfitting risks. A global affinity regression block receives extracted features before producing a binding affinity score through its final prediction. The design of DeepAtom achieves both model complexity and generalization because it enables effective performance when trained with limited labeled data. Leaky ReLU activation functions together with batch normalization enhance the computational speed while using training processes to achieve more stable convergence results. DeepAtom was tested and evaluated thoroughly using PDBbind v.2016 along with the Astex Diverse Set. DeepAtom was also trained using PDBbind's refined and general subsets with data augmentation by applying random translations and rotations to develop robustness. According to the evaluation, DeepAtom outperformed the well-known RF-Score, Pafnucy, and X-Score models. DeepAtom demonstrated outstanding performance on PDBbind v.2016 core set with R of 0.83 and RMSE of 1.23 pK units better than other prevailing methods. The model received enhancement through creation of a benchmark dataset that served both as a performance metric and useful resource for future studies in computational drug discovery.

Suruliandi et al. (2024)[27] explore DTI prediction through ML techniques to strengthen drug discovery operations. The research analyzes multiple machine learning (ML) algorithms for DTI expectation while emphasizing the significance of detecting potent drug-protein interactions to develop less expensive new medications. The researchers demonstrate the effectiveness of ML technique predictions for drug and protein target interactions through their analysis of diverse classifiers and datasets and qualitative and quantitative data assessment. The authors established that multiple ML approaches lead to promising DTI prediction results, with chemogenomics-based methods showing the most potential. The research team investigated three major ML strategies that address different functions during DTI prediction, including binding site identification and analysis of drug and target chemical-genomic spaces through docking-based, ligand-based, and chemogenomics-based approaches. Future research in this field should develop new classifiers to improve prediction accuracy according to the study, which reports minimal accuracy performance because of limited true negative drug-target pair availability. The paper stresses the necessity to advance ML techniques handling extensive data quantities as well as the integration of continuously updating data sources to optimize prediction procedures. Scientists plan to develop both negative datasets for interaction predictions in addition to applying feature engineering techniques for enhancing dataset quality in order to achieve reliable DTI prediction through advanced ML models.

Zhou et al. (2024)[32] introduce a solution to improve accurate drug-protein interaction (DPI) predictions that serve important functions in drug discovery and personalized medicine. Deep learning models currently face challenges when representing nodes accurately since this issue leads to reduced predictive outcomes. The authors developed a new framework that combines global and local node characteristics from drug-protein bipartite graph systems. The method employs pre-trained extraction of initial features for drugs along with proteins. Local estimations occur through MinHash with HyperLogLog that measure subgraph similarity and set cardinality together with a transformer-based global diffusion mechanism that monitors node interdependencies. The final part merges local components with global elements before applying a multilayer perceptron (MLP) to determine DPI probability. The model shows high accuracy levels through various experimental validations which prove it predicts drug-protein interactions better than conventional techniques. Molecular docking experiments validate that the model identifies fresh DPIs which were absent from existing databases therefore showing potential for drug rejuvenation purposes. Additionally the authors note challenges in managing complex data sources as well as difficulties in making their predictions easier to understand. The future development plan includes integrating new data types alongside enhanced interpretability mechanisms alongside framework optimization for enhanced prediction accuracy.

Bal et al. (2024)[24] used their research to solve the prediction accuracy problem for drug-target interactions (DTIs), which are essential for drug discovery, by developing better computational models for drug-protein binding affinity measurements. Binary classification methods used traditionally do not represent the continuous characteristics of binding strength. The authors present PGraphDTA as their solution to this limitation through their combination of Protein Language Models (PLMs) with Contact Map information. The implementation of ProtBERT, DistilProtBERT, ESM2, and SeqVec serves as Protein Language Models to produce strong representations of amino acid sequences instead of using previous Convolutional Neural Networks (CNNs). The predictive framework includes contact maps as inductive biases which provide protein residue-based and molecular distance-based components to enhance binding affinity predictions. The investigation shows how PLMs enhance prediction accuracy by using DAVIS and KIBA datasets where SeqVec delivers the minimum mean squared error (MSE). The performance benefit from protein contact maps in systems outweighs the negative effects of molecular docking-based contact maps that introduce errors into results. The research shows how PLMs boost protein data representation, especially for short datasets, and contact maps generate valuable structural information. The main challenges of this work include the high processing cost of embedding generation using PLMs along with uncertainties coming from molecular docking predictions. The ongoing work aims to merge physical element-based structural biases together with attention systems to enhance interaction design while conducting tests through larger available data collections. The study has developed an essential framework to improve DTI prediction by creating better computation methods, which will boost drug discovery speed.

Kawichai et al. (2021)[13] provide an ensemble learning approach as a means of predicting drug-disease relationships. Drug repositioning issues may be resolved using Meta-path-based Gene Ontology Profiles for Drug-Disease Associations (MGP-DDA). Drug development through traditional techniques takes an extended period with high costs, which leads research to explore computational approaches that find additional uses for existing medications. The authors developed a meta-path methodology that incorporates Gene Ontology (GO) terms as connecting nodes between drug and disease entities to generate new information features named meta-path-based GO profiles. The profiles apply drug-disease functional similarities to provide accurate predictions about drug-disease associations. The research design builds a three-part network that relies on information from DrugBank together with DisGeNET and the Gene Ontology Annotation (GOA) database. The meta-paths drug-GO-disease and drug-GO-drug-disease and drug-disease-GO-disease lead to the creation of GO profiles that undergo singular value decomposition (SVD) for dimensionality lowering. The combination of PU bagging with XGBoost as its base learner trains the features to produce predictions. The experimental study demonstrates that MGP-DDA produces superior performance to current methods by reaching 88.6% precision. The practical benefits of MGP-DDA become evident through the fact that 37.7% of predicted drug-disease associations receive validation in ClinicalTrials.gov. The research identifies processing expenses as well as the requirement for incorporating more medical attributes into the system as current limitations. The upcoming research efforts will work on making the results more understandable and include pathway data to enhance predictive capabilities.

Zhang et al. (2024)[30] tackle drug-protein interaction (DPI) prediction challenges because they are essential for drug discovery despite the prevalent bias in graph neural networks (GNNs) and negative transfer across bipartite networks. The authors present Meta-Graph Association-Aware Contrastive Learning (MGACL) to combine meta-knowledge transfer methods with contrastive learning for improving DPI prediction accuracy. A heterogeneous graph neural network within MGACL evaluates complex drug–target relationships through the combination with meta-local connected knowledge transfer networks for improved drug- and target-specific representation enhancement. The model uses an association-aware contrastive learning method that connects high-frequency drug vectors with learned auxiliary graph vectors to stop the negative transfer from occurring. The predictions of MGACL outperform existing approaches by approximately 3% according to AUC and AUPRC metrics through evaluations conducted on five different datasets. The model's effectiveness proves itself even more through confirmations from molecular docking testing of predicted drug–target connections. The research notes two weaknesses, which include difficulty processing uncommon drugs along with expensive training costs for large heterogeneous networks and the existence of improved yet observable negative transfer. Researchers will conduct future work to improve powerful representation interpretability methods while also optimizing contrastive learning techniques and integration of features to enhance DPI prediction accuracy results.

Kyro et al. (2023)[22] explore the prediction of protein-ligand binding affinity because this ability drives drug discovery and protein engineering research. The authors propose HAC-Net as a Hybrid Attention-Based Convolutional Neural Network which combines 3D convolutional neural network functionality with two graph convolutional networks using attention techniques to boost prediction accuracy. Heatmap-based 3D-CNN deals with atomic-scale spatial data and incorporates GCNs to treat molecular connections as graph structures and applies node selection methods for feature aggregation. HAC-Net attains the top performance rating for PDBbind v.2016 core set benchmarks because of its integrated approach. The model's generalization ability is tested using three sets of training-test separations that optimize differences between protein structures and sequences together with ligand fingerprint variations. HAC-Net's practicality is confirmed when tested on lower-quality data in addition to having its robustness validated through ten-fold cross-validation procedures for ligand variation. HAC-Net achieves superior outcomes compared to prior models by delivering better results in root-mean-square error (RMSE) Spearman rank correlation and Pearson correlation measurement. The use of attention-based components leads to better prediction results because they help identify new interactions between proteins and ligands. The study authors identify two main disadvantages in their approach, such as deep learning network training expenses and potential performance biases in benchmarked data. The research field intends to maintain representation interpretability while optimizing attention elements for use across additional biomolecular property prediction responsibilities. HAC-Net exists on both GitHub and PyPI repositories for scientific community members to develop and validate its use through standardized frameworks.

Gao et al. (2024) [34] focused on the difficulties related to the prediction of Adverse Drug Reactions (ADRs), which is among the most important problems, with a goal in mind for an improvement of both patient's safety and healthcare resources consumption. Classical ADR prediction models tend to not easily handle complex properties involving individual patients due to parameter-dependent model foundations particularly characterised by demographic differences (e.g., age and sex), which also significantly affect the existence of ADRs in a given setting. To close this gap, the authors presented PreciseADR, a system with the goal of personalized ADR prediction at individual patient level. The model includes information from three sources which are categorised into demographic and clinical information for a new patient representation. By running GNNs, particularly the HGNN model, they model correlation relationships among patients, drugs, diseases and ADRs as nodes in a heterogeneous graph and link them with edges. Experiments reveal significant improvement in the prediction performance achieved by PreciseADR over the state of art machine learning methods, which achieves 3.2% AUC and 4.9% Hit@10 improvements respectively. These results corroborate the model's capability of learning intricate dependencies in patient data. In addition, we integrate demographic characteristics such as age and gender, showing they have a notable impact on ADR prediction and the potential of our method in advancing precision medication. Nonetheless, despite being able to incorporate multiple clinical information, the model did not include omics data that could limit its ability to capture biological heterogeneity. Therefore, the authors suggest including omics-level data in future studies as well as advanced calculation data augmentation methods to increase predictive stability of ADR models.

Li et al. (2025) [25] aimed at mitigating the prediction of Adverse Drug Reactions (ADRs), one of the most persistent issues in global healthcare, owing to the tremendous effects they have on both patient morbidity and mortality. Conventional ADR prediction models usually encounter a problem of sparse and high-dimensional feature data, which not only deteriorates the prediction accuracy but also leads to training an individual model for each ADR. In order to tackle these shortcomings, the authors proposed a novel model that can address both KG embeddings and deep learning for accurate ADR prediction. In such a way, drug-related objects (i.e., enzymes, drugs, and ADRs) can be transformed into low-dimensional continuous vector spaces through employing the KGs for addressing the data sparsity problem while representing complicated biological essence. Moreover, their deep learning method and integrative features effectively learn the massive amount of data to improve the prediction significantly. Mixing the technology of KG embeddings and CNNs, our model provides unified prediction over multiple ADRs without constructing separate classifiers, which can be expensive in computational costs and challenging to scale. Experimental results suggest that our model is able to learn from the graph structure and outperform traditional machine learning algorithms, which are not powerful enough to catch the complicated relationship among drugs, enzymes, and side effects. The predictive robustness in the present study is also strengthened by the integration of a variety of features, including drug transporters and therapeutic indications. Yet, the paper indicates that there is a lack of standard benchmarking datasets, and the design of optimized embedding strategies are still an open question. In spite of these limitations, Li et al. 's method is a good step forward in developing more generalizable and efficient ADR prediction models.

Zhang et al. (2021) [17] addressed a major bottleneck, ADR prediction, which still represents one of the most crucial issues in drug research at present since today's clinical trials and animal testing fail to detect any potential ADRs before drugs can be put on the market. Given that these traditional approaches cannot fully process the complicated drug-drug interaction and handle a great diversity of compounds, they proposed a novel computational framework by integrating data mining and machine learning methods for more accurate ADR prediction. They were the first to suggest an approach based on KG embedding, with their method representing complex relationships between drugs, side effects, targets and indications as distributional embeddings. It allows the model to learn not only direct ADR hypotheses but also indirect connections between biomedical entities, facilitating a comprehensive ADR identification. To predict the ADRs better, we also integrated KG embeddings with logistic regression to form a general prediction model without requiring different models built for different ADRs. The approaches were tested on benchmark ADR datasets and showed a remarkable performance with high AUC up to 0.87, compared with other convolutional neural networks (CNNs) or deep heterogeneous joint models that tend to be fragile to model's interpretation or computationally expensive. The outcomes demonstrate that the KG embeddings can help to provide more accurate and interpretable ADR predictions. However, Zhang et al. (2021) also noted that the sparsity of data when applied to achieve improved predictive performances, and scalability for large-scale biomedical research, remains a challenge. The support of new research to develop better semantic drug-disease representation and the implementation of an ensemble of DNN-based framework 32 are needed to improve both the accuracy and generalization of ADR prediction.

Park et al. (2022) [18] against the long-standing challenge of ADR prediction (an issue continuing to hugely impede drug development due to their complex off-target interactions). In vitro experiments and the molecular docking simulations are conventional means for studying drug–target interactions; however, they can only perturbate known proteins, which decreases their predictive scope and efficacy. To overcome these shortcomings, in this study the authors proposed a computational model named as LAPINE (Large-scale ADR-related Proteins Identification with Network Embedding), which was created based on a network approach to enhance the identification accuracy of ADR-related proteins. LAPINE In this paper, combines the data of single-target compounds (STCs) through network embedding algorithms to capture more intricate relationships between drugs with targets/proteins in a higher systemic level. Importantly, it encompasses not only DPIs but also protein–protein interactions (PPIs) and could provide a more comprehensive description in putative ADR-related pathways. By doing so, such a network embedding can encode the biological relatedness in multi-dimensions, in which the representation can reflect not only local but also global topological features. Performance assessment also showed that LAPINE could achieve higher values of PPV and LR than traditional models built on drug intensities-target interactions alone across several large benchmarks of known ADR-related proteins. In Figure 7, results supported that the model can have better ability in providing predictions of more novel ADR-related proteins, which many of them were then verified by literature evidence (Galletti et al., 2022). By contrast to the previous models based on drug-only methods, LAPINE showed a better tradeoff between precision and coverage, which made it stand out for practical use in pharmacovigilance studies. Despite the good performance of this study, it also has some limitations. The performance of the model and subsequently its demonstrated validity to predict certain processes is somewhat contingent on the quality and comprehensiveness of interaction networks, which may introduce biases; additionally, other molecular or omics data could increase consideration. The authors propose the application of multi-modal data fusion and deep graph learning approaches to enhance ADR prediction accuracy as well as interpretability.

2.3 Summary of Key Findings

2.3.1 Transition from handcrafted to learned representations

The literature clearly evidences that traditional machine learning algorithms based on manually engineered molecular descriptors are being replaced by deep learning methods, which can learn data-driven features. State-of-the-art CNN- and transformer-based models such as ChemBERTa, DeepDTA[paper no 1], and GraphDTA[Paper no 16] overcome prior methods by learning to automatically mine structural/chemical patterns directly from raw inputs in order to generalize more effectively across wide ranges of chemical/molecular classes.

2.3.2 Emergence of structural and multi-modal modeling

New studies have pointed out the increasing roles of 3D structure for drugs and proteins as well. For GVP-GNN and EGNN, which are based on graph, geometry and topology cues from the real molecular system can probably be expressed more faithfully. This transformation is the transition from structure-agnostic to structure-aware learning in which molecular geometry plays a central role for optimal prediction.

2.3.3 Integration of Pretrained Foundation Models

The use of large-scale pretrained models, like ChemBERTa in the domain of chemicals and ESM in the case of proteins, has made itself a common practice for representation learning. Their contextualized embeddings encode representations that contain rich biochemical information, and fine-tuning them on DTI-related problems leads to better performance and interpretability. This trend highlights a general adoption of foundation models in bioinformatics, where pretrained networks become modularized building blocks for specialized downstream tasks.

2.3.4 The Coming of Contrastive and Multi-Task Learning

New literature increasingly provides evidence that drug behavior cannot be explained using a single criterion. Multi-task learning models that optimize related pharmacological endpoints-like DTI and ADR-enable a joint learning among the molecular and clinical spaces. The contrastive learning, on the other hand, enhances representation alignment in disparate modalities such that three sets of drug, protein, and ADR embedding occupy meaningful areas in a common hidden space.

2.3.5 Rise of contrastive and multi-task learning

Modern research is more and more aware that the action of a drug can not be modeled with just a single objective. Due to anatomical, pharmacological, and clinical correlations (DTI vs ADR) in related aspects of the human brain and body, multi-task architectures that jointly optimize multiple related pharmacological outcomes —such as DTI and ADR enabled shared learning between molecular and clinical modalities. On the other hand, contrastive learning facilitates representation alignment of heterogeneous modalities such that drug, protein and ADR embeddings are located close to each other in the latent space.

2.3.6 Need for Unified DTI–ADR Frameworks

Despite these achievements, most previous studies treat DTI and ADR prediction independently. This disconnection ignores their biological crosstalk, among which off-target binding is a frequent cause of side reactions. The set of reviewed papers taken as a whole seem to support the need for an integrated multi-task framework that is capable of modeling both therapeutic and side effects at the same time. Such integration may also establish a missing link between molecular-level prediction and patient-level outcomes, thus improving real-world drug safety and discovery pipelines.

Chapter 3

Proposed Methodology

3.1 Design Process or Methodology Overview

This research framework was designed because the aim was to develop a single deep learning system that can predict both drug-target interactions (DTI) and adverse drug reactions (ADRs) in the same architecture. This two-fold prediction objective demanded a compromise between two different yet biologically connected issues: classification of molecular binders and clinical outcome regression. The main methodological issue was to incorporate heterogeneous data sources chemical sequences, molecular graphs, protein sequences and three-dimensional structures into a consistent format of computation and to make sure that each modality is an important factor in the learning process.

In order to meet these needs, a multi-task architecture with a modular design was created. The system uses a common space of representations which aligns drugs, proteins and ADRs, using small networks of adapters. The adapters stabilize the variability of scale and dimension of the types of embedding and project all the inputs into a 512-dimensional latent space. This modularity allowed the architecture to flexibly replace all combinations of different encoders without redesigning the downstream prediction heads so as to have a reasonable comparative evaluation across a range of experiments.

The design process starts with baseline experiments using single-encoder embeddings. The modality was tested individually in order to learn the discriminative power of each representation. In the case of proteins, the Evolutionary Scale Model (ESM) [1] and Geometric Vector Perceptron (GVP) [12] embedding types were compared, whereas in the case of drugs, ChemBERTa [7], Smiles2Vec [2], and E(n)-Equivariant Graph Neural Network (EGNN) [35] types of embedding were analyzed. Of these, ESM of proteins and Smiles2Vec of drugs were most stable, and interpretable, which reports that sequence-based encoders can be used alone to learn a lot about biochemical properties.

Nonetheless, single embeddings were found to be limited in the representation of both chemical and structural geometry of molecules. To surmount this, the second step involved fusion experiments where complementary embeddings were appended prior to entering the adapter layers. Several combinations were tried, although only two combinations frequently performed better than all other combinations:

- ESM + GVP for proteins, and
- ChemBERTa + EGNN for drugs.

These examples demonstrated that the sequence-based embeddings are able to capture semantic and evolutionary features, whereas structure-based embeddings can give geometric and spatial context; a combination of the two makes richer representations that are better generalized across unseen drug-protein pairs.

These examples were improved, and this confirms that sequence-based embeddings have the ability to capture semantic and evolutionary information, but structure-based embeddings have access to geometric and spatial context; a combination of the two can create richer representations that are capable of generalizing across unseen pairs of drugs and proteins.

In addition to representation learning, the model utilized contrastive alignment of drug-protein and drug-ADR space. This auxiliary goal stimulated the integration of biologically related objects into surrounding areas in the shared space, despite the fact that explicit supervision was restricted. Supervised losses (to predict DTI and ADR) were combined and so offset local discrimination versus global consistency in the learned manifold.

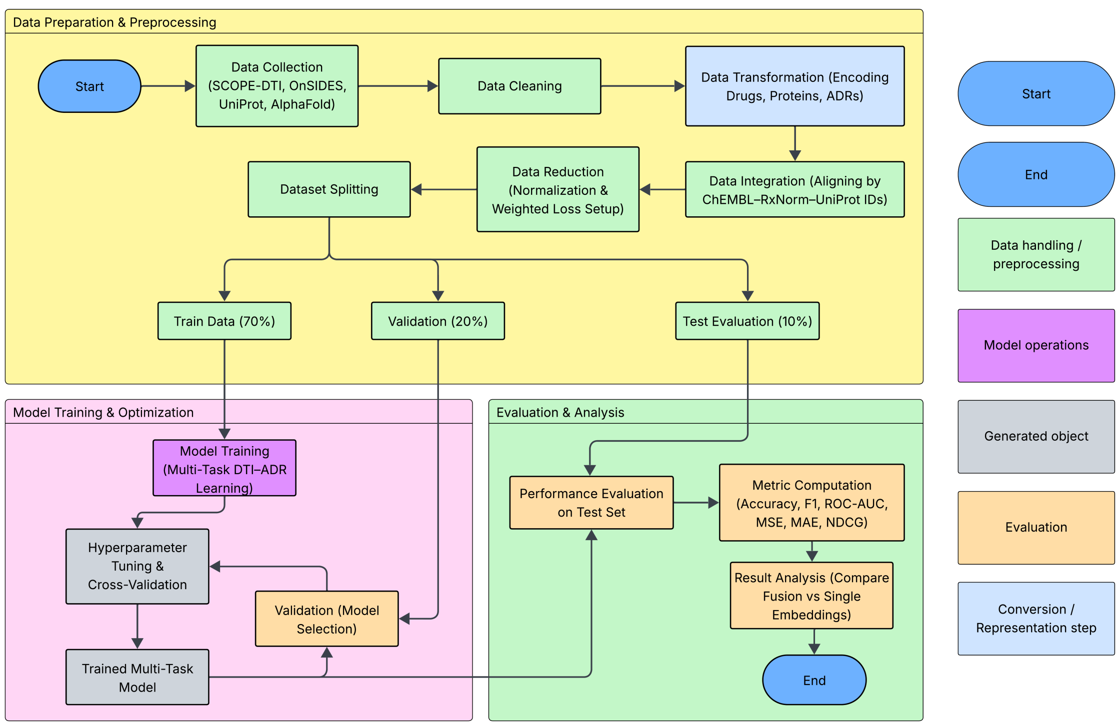

Figure 3.1: Workflow diagram

In short, the design process evolved through an experimental process- alternating single-embedding baselines to fused multimodal representations incorporated in a standard adapter-based multi-task architecture. Empirical studies revealed that the fusion-based model using ESM + GVP (protein) and ChemBERTa + EGNN (drug) could achieve maximum trade-off between the predictive accuracy, generalization, and interpretability. This model therefore became the final design that would be used and discussed further in the subsequent sections.

3.2 Model Specification

The suggested architecture deploys a multi-task deep learning system that collaboratively predicts drug-target interactions (DTIs) and adverse drug reactions (ADRs). The model conditions a mutual biochemical representation space, which aids in making two prediction heads:

- A DTI head determining binding probability between a protein and drug.
- An ADR head predicting the continuous TF-IDF-weighted association profile of adverse effects for that drug.

This section presents the formal specification of each module and the complete training objective.

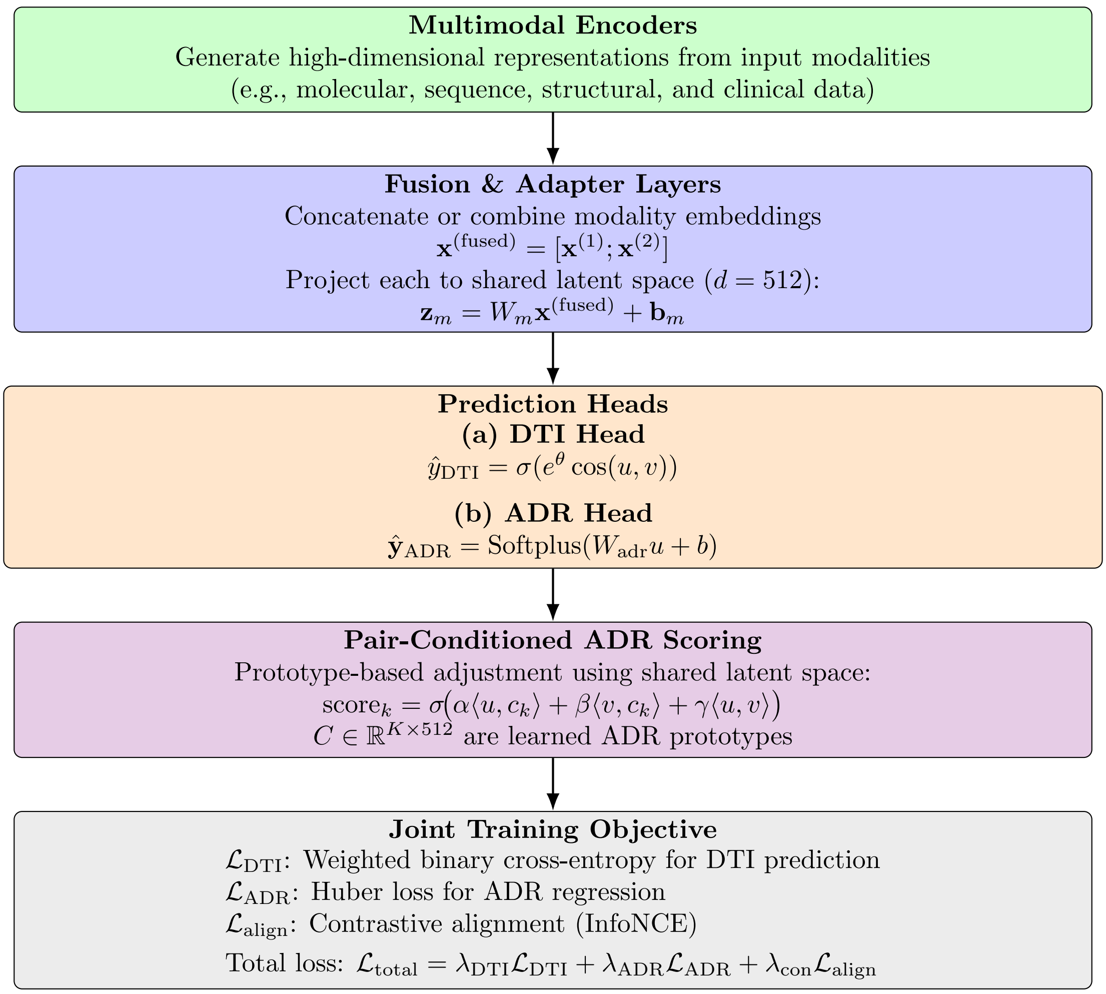

Figure 3.2: Generalized model architecture

3.2.1 Input Representations

The heterogeneous biochemical representations ingested by the model include drug molecules, protein targets and adverse drug reactions (ADRs). Each entity type originates from different encoders, yielding features vectors of varying sizes and statistical distributions. To ensure consistent mathematical treatment, these heterogeneous embeddings are formalized as follows.

Mathematical Formulation

- $\mathbf{x}_d \in \mathbb{R}^{n_d}$: Drug embedding vector, derived from ChemBERTa and EGNN encoders;
- $\mathbf{x}_p \in \mathbb{R}^{n_p}$: Protein embedding vector, derived from ESM and GVP encoders;
- $\mathbf{t}_d \in \mathbb{R}^{K}$: TF–IDF vector of ADR associations, where $K = 4048$.

For the fused configuration used in the final model:

$$n_d = 384 + 256 = 640, \quad n_p = 1280 + 1024 = 2304.$$

Each drug–protein pair (d, p) from the dataset is coupled by a binary DTI label $y_{\text{dti}} \in \{0, 1\}$ and an ADR target vector $\mathbf{t}_d$.

Dataset Composition

- $N_{\text{pairs}} = 34{,}741$ drug–protein pairs (12,234 positive interactions, 22,507 negative).
- $N_{\text{drugs}} = 1{,}028$, $N_{\text{proteins}} = 2{,}385$.
- ADR feature space dimensionality $K = 4048$.

The IDF constants estimated were used to create the ADR TF-IDF matrices just on the training drugs, so that there is no information leakage across validation or test. sets. All the elements of $\mathbf{t}_d$ are weighted frequencies of ADR terms related to that drug based on the MedDRA vocabulary.

Fusion of Embeddings

Before entering the adapter these embedding are concatenated, in order to capture both semantic and geometric information:

$$\mathbf{x}_d^{(\text{fused})} = \left[\mathbf{x}_d^{(\text{ChemBERTa})};\ \mathbf{x}_d^{(\text{EGNN})} \right] \in \mathbb{R}^{640},$$
$$\mathbf{x}_p^{(\text{fused})} = \left[\mathbf{x}_p^{(\text{ESM})};\ \mathbf{x}_p^{(\text{GVP})} \right] \in \mathbb{R}^{2304}.$$

This concatenation operation preserves the complete feature information from both modalities without introducing additional transformation bias prior to shared-space projection.

3.2.2 Adapter Networks for Shared Space Learning

The drug, protein encoders, and ADR embeddings are based on different encoders that have different dimensions and statistical scales. To have significant joint learning, these representations have to be converted into a shared latent space in which distance, similarity, direction have analogous geometrical significance. This is accomplished via three adapter networks, referred to a f_d, f_p, f_a, representing drugs, proteins and ADRs respectively.

All the adapters carry out (i) dimensional alignment, (ii) normalization, and (iii) mild nonlinear transformation to enhance representational expressiveness with efficiency. Each of the adapters is a mapping of their input vectors to a common embedding dimension $d = 512$.

Mathematical Formulation

Let the three adapters be defined as:

$$\begin{aligned}
\mathbf{u} &= f_d(\mathbf{x}_d^{(\text{fused})}) \in \mathbb{R}^d, \\
\mathbf{v} &= f_p(\mathbf{x}_p^{(\text{fused})}) \in \mathbb{R}^d, \\
\mathbf{w} &= f_a(\mathbf{t}_d) \in \mathbb{R}^d.
\end{aligned}$$

Each adapter network follows the same architecture pattern:

$$f(\mathbf{x}) = \text{LayerNorm}\left(W_2\,\text{Dropout}\left(\text{GELU}(W_1\mathbf{x})\right)\right),$$

where $W_1 \in \mathbb{R}^{(2d)\times n_{\text{in}}}$ and $W_2 \in \mathbb{R}^{d\times(2d)}$.

This two-layer projection first expands dimension of the feature by a factor of 2, then to introduce non-linearity it applies Gaussian Error Linear Unit (GELU), and finally projects back to the target size d. The layer normalization standardizes feature statistics so that modalities have similar scales, which is very important in cosine-similarity-based contrastive training.

To reduce overfitting to infrequent ADR patterns with ADR vectors that are typically high-dimensional and sparse, an input dropout of 0.1-.2 is used prior to the initial linear transformation.

Dimensional Mapping

$$\begin{aligned}
f_d &: \mathbb{R}^{640} \longrightarrow \mathbb{R}^{512}, \\
f_p &: \mathbb{R}^{2304} \longrightarrow \mathbb{R}^{512}, \\
f_a &: \mathbb{R}^{4048} \longrightarrow \mathbb{R}^{512}.
\end{aligned}$$

Once this transformation has been made, the entity types are all in the same vector space, so that similarities can be directly computed, objective comparisons can be made, and multi-task learning can be shared.

Adapter Output Geometry

The resulting vectors $(\mathbf{u}, \mathbf{v}, \mathbf{w})$ form the backbone of the model's geometric reasoning:

- $\mathbf{u}$: latent representation of a drug
- $\mathbf{v}$: latent representation of a protein
- $\mathbf{w}$: latent representation of ADR profile

These vectors are then computed in terms of their spatial relationships to compute drug-protein binding probabilities, align drug-ADR semantics, and even pair-specific ADR scoring.

3.2.3 DTI Head – Binding Probability Estimation

Drug-Target Interaction (DTI) head is tasked with the responsibility to predict the likelihood of a drug molecule to bind to a certain protein target.The DTI head compares the two embeddings into the shared latent space in terms of respective adapters, and the geometric compatibility between them is estimated with respect to temperature-scaled cosine similarity. This design allows the model to be trained to the optimal sensitivity of similarity scores to the distribution of interaction labels and is also resistant to the scale of features.

Mathematical Formulation

Where $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$ are the latent embeddings of a drug and a protein respectively, which are obtained through their adapter f_d, f_p.The raw interaction logit is specified as:

$$z = s \cdot \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\| + \varepsilon},$$

where:

- $\mathbf{u} \cdot \mathbf{v}$ is the dot product,
- $\|\mathbf{u}\|$ and $\|\mathbf{v}\|$ are the ℓ_2 norms,
- ε is a small constant (e.g., 10^{-8}) to prevent division by zero, and
- $s = \exp(\theta)$ s a learnable positive scale parameter (initialized around 10.0), in order to allow the model to adjust similarity magnitudes.

In order to get the predicted value, the logit is passed through a sigmoid function to get the predicted interaction probability:

$$\hat{y}_{\text{dti}} = \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Therefore, the projection of the shared-space representations to the likelihood of interaction can be summed up as follows:

$$\hat{y}_{\text{dti}} = \sigma \left(\exp(\theta) \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\| + \varepsilon} \right).$$

Loss Function

Since DTI dataset is skewed ($N_{\text{pos}} = 12{,}234$, $N_{\text{neg}} = 22{,}507$), to prevent the imbalance in the training objective, the weighted binary cross-entropy (BCE) is used to attribute higher weight to the positive samples:

$$\mathcal{L}_{\text{dti}} = -\frac{1}{N}\sum_{i=1}^{N}\left[w^{(+)}y_i \log \hat{y}_i + (1-y_i)\log(1-\hat{y}_i)\right],$$

where $w^{(+)} = \frac{N_{\text{neg}}}{N_{\text{pos}}} \approx 1.84$.
Such weighting allows the model not to degenerate into a trivial non-binding predictor and stabilizes knowledge in skewed label ratios.

Optimization Details

The logits z_i are calculated on a mini-batch of drug protein pairs during training and fed through `torch.nn.BCEWithLogitsLoss(pos_weight=w_pos)` in the implementation. The parameter to be learned θ is optimized with the rest of the model through AdamW optimization.

Geometric Interpretation

The embeddings of each drug and protein can be considered a point on a hypersphere in the common latent space. A positive interaction is associated with a small angular distance (large cosine similarity) and a non-interaction is associated with near-orthogonality. The learnable scale s a temperature that regulates the sharpness with which the model is distinguishing between close and distant pairs.

3.2.4 ADR Head – TF-IDF Regression

The Adverse Drug Reaction (ADR) head estimates the strength of the relationship between a drug and each of the $K = 4048$ terms in the TF-IDF feature space that are represented by ADRs. Contrary to the DTI head, which relies on the drug and protein embeddings, the ADR head relies on the drug representation $\mathbf{u}$ in the shared latent space only. . The decoupling also enables the model to learn in order to learn a drug-centric embedding augmented with pharmacological environment, which is further applied during pair-specific ADR ranking.

Mathematical Formulation

Given a drug d and latent embedding $\mathbf{u} \in \mathbb{R}^d$, the ADR head uses a linear transformation, followed by a Softplus non-linearity to make all the outputs non-negative:

$$\hat{\mathbf{y}}_{\text{adr}} = \text{Softplus}(W_{\text{adr}}\,\mathbf{u} + \mathbf{b}_{\text{adr}}), \quad W_{\text{adr}} \in \mathbb{R}^{K\times d},\ \mathbf{b}_{\text{adr}} \in \mathbb{R}^{K}.$$

The outcome $\hat{\mathbf{y}}_{\text{adr}} \in \mathbb{R}^K$ is the approximation of the target TF-IDF vector $\mathbf{t}_d$ and each component $\hat{y}_{\text{adr},k}$ predicts the relative likelihood or strength of ADR k for that drug.

The Softplus activation

$$\text{Softplus}(x) = \ln(1 + e^x)$$

maintains gradient smoothness and outputs are limited to non-negative which is compatible with TF-IDF weighting.

Huber Loss Objective

The mean Huber loss between predicted and actual TF-IDF values are minimized by training:

$$\mathcal{L}_{\text{adr}} = \frac{1}{K} \sum_{k=1}^{K} \text{Huber}\left(\hat{y}_{\text{adr},k},\ t_{d,k}\right),$$

where for residual $r_k = \hat{y}_{\text{adr},k} - t_{d,k}$,

$$\text{Huber}(r_k) = \begin{cases} \frac{1}{2} r_k^2, & |r_k| \le \delta, \\ \delta\left(|r_k| - \frac{1}{2}\delta\right), & |r_k| > \delta, \end{cases} \qquad \delta = 1.0.$$

The Huber loss is quadratic on small errors and linear on large ones, minimizing the impact of the outlier ADR weights that are caused by the occurrence of rare but heavily weighted TF-IDF terms. This powerful formulation can be used to stabilize the training due to the sparseness and heavy-tailed nature of the ADR values.

Interpretation

The ADR head is successful in recreating the ADR profile of every drug in a regression-like fashion. The latent drug embedding $\mathbf{u}$ will represent a compression of a pharmacological signature of both molecular and side-effect semantics. This is a vector that subsequently reacts with protein embeddings via the pair conditioned ADR scoring mechanism in Section 3.2.6.

3.2.5 Contrastive Learning Objectives

Even though the supervised DTI and ADR losses give the model explicit task constraints, it is not known that the latent representation $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are a semantically consistent geometric space.This model uses contrastive learning to bring related entities closer and unrelated ones farther apart to provide this structure. In particular, there are two opposing contrasting goals presented:

- Drug–Protein contrast ($\mathcal{L}_{dp}$) — Promotes embeddings of binding pairs to be near each other.
- Drug–ADR contrast ($\mathcal{L}_{da}$) — promotes drug embeddings matching their own ADR profiles but distanced to unrelated ADRs.

Both losses use InfoNCE formulation which is a normalized cross-entropy with in-batch negatives.

Drug–Protein Contrast

To select a mini-batch of B positive drug protein pairs $(\mathbf{u}_i, \mathbf{v}_i)$, every pair of them is regarded as a positive match, and all the rest of the pairings in the batch are regarded as negatives. Embeddings are ℓ_2-normalized:

$$\tilde{\mathbf{u}}_i = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}, \quad \tilde{\mathbf{v}}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}.$$

The pairwise similarity matrix is then computed as

$$S_{ij} = \frac{\tilde{\mathbf{u}}_i \cdot \tilde{\mathbf{v}}_j}{\tau},$$

with $\tau > 0$ a temperature hyper-parameter that regulates the sharpness of the distribution ($\tau = 0.07$ in this work).

The drug to protein direction InfoNCE loss is

$$\mathcal{L}_{dp}^{u\to p} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(S_{ii})}{\sum_{j=1}^{B} \exp(S_{ij})}.$$

A symmetric counterpart for protein-to-drug alignment is defined as

$$\mathcal{L}_{dp}^{p\to u} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(S_{ii})}{\sum_{j=1}^{B} \exp(S_{ji})}.$$

The combined Drug–Protein contrastive loss becomes

$$\mathcal{L}_{dp} = \frac{1}{2}\left(\mathcal{L}_{dp}^{u\to p} + \mathcal{L}_{dp}^{p\to u}\right).$$

This promotes the occupancy of a high similarity neighborhood by each binding drug-protein pair and repulsion of non-binding pairs.

Drug–ADR Contrast

A parallel formulation is used to align every drug embedding $\mathbf{u}_i$ with its adapted ADR representation $\mathbf{w}_i = f_a(\mathbf{t}_{d_i})$. After normalization:

$$\tilde{\mathbf{w}}_i = \frac{\mathbf{w}_i}{\|\mathbf{w}_i\|},$$

the similarity matrix is computed as

$$S_{ij}^{(da)} = \frac{\tilde{\mathbf{u}}_i \cdot \tilde{\mathbf{w}}_j}{\tau}.$$

The bi-directional InfoNCE objective is analogous:

$$\mathcal{L}_{da}^{u\to a} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(S_{ii}^{(da)})}{\sum_{j=1}^{B} \exp(S_{ij}^{(da)})},$$

$$\mathcal{L}_{da}^{a\to u} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp(S_{ii}^{(da)})}{\sum_{j=1}^{B} \exp(S_{ji}^{(da)})},$$

$$\mathcal{L}_{da} = \frac{1}{2}\left(\mathcal{L}_{da}^{u\to a} + \mathcal{L}_{da}^{a\to u}\right).$$

The joint contrastive term enforces the fact that every drug vector $\mathbf{u}_i$ i is close to its own ADR vector$\mathbf{w}_i$ and far away that of other drugs. Such implicit supervision is used to transfer pharmacological semantics to the DTI representation in the case of sparse ADR annotations.

Total Contrastive Loss

The last contrastive element is the total of the two terms of alignment:

$$\mathcal{L}_{\text{contrast}} = \mathcal{L}_{dp} + \mathcal{L}_{da}.$$

It is used together with the supervised losses in the process of optimization as weighted auxiliary guidance (see Section 3.2.8).

3.2.6 Pair-Conditioned ADR Scoring

Even though ADR labels can only be obtained at the drug level, the model can be used to estimate the pair-specific ADR risks by conditioning on the latent representation of the protein. This is done by using a prototype based inference mechanism which learns the general embedding pattern of the ADRs in the shared space and compares it to the drug embedding, as well as protein embedding

Conceptual Overview

Once the model is trained using DTI and ADR supervision, each of the ADR types (e.g. headache, nausea, rash) is represented by a prototype vector that summarizes the average embedding of all the drugs known to exhibit that ADR. The model at the inference time will compare the embedding of a drug $\mathbf{u}$ and a protein $\mathbf{v}$ to these ADR prototypes $\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_K]^\top$. The resultant similarity scores give a ranked ADR list particular to the (drug, protein) pair being tested.

Construction of ADR Prototypes

For each ADR $k \in \{1, \ldots, K\}$, where $\mathcal{D}_k$ as the set of drugs with non-zero TF-IDF value of such ADR. The prototype, $\mathbf{c}_k$ is defined as the mean of the adapted ADR embeddings $f_a(\mathbf{t}_d)$ for all drugs in that set:

$$\mathbf{c}_k = \begin{cases} \frac{1}{|\mathcal{D}_k|} \sum_{d \in \mathcal{D}_k} f_a(\mathbf{t}_d), & \text{if } |\mathcal{D}_k| > 0, \\ \mathbf{0}, & \text{otherwise.} \end{cases}$$

The resulting prototype matrix

$$\mathbf{C} = [\mathbf{c}_1, \mathbf{c}_2, \ldots, \mathbf{c}_K]^\top \in \mathbb{R}^{K \times d}$$

is fitted to the training drugs and is held constant at inference. The prototypes serve as centroids of latent space of each type of ADR.

Scoring ADRs for a Drug–Protein Pair

Given an embedding of a drug $\mathbf{u} \in \mathbb{R}^d$ and a protein embedding $\mathbf{v} \in \mathbb{R}^d$, the model computes a pair-conditioned ADR score, score$_k$, for every ADR prototype $\mathbf{c}_k$ in the following way:

$$\text{score}_k = \sigma\left(\alpha\left(\mathbf{u} \cdot \mathbf{c}_k\right) + \beta\left(\mathbf{v} \cdot \mathbf{c}_k\right) + \gamma\left(\mathbf{u} \cdot \mathbf{v}\right)\right),$$

with $\sigma(\cdot)$ the sigmoid activation and the learnable scalar weights α, β, γ controlling the contribution of drug–ADR, protein–ADR and drug–protein similarity respectively.

The term $(\mathbf{u} \cdot \mathbf{c}_k)$ measures the similarity between the drug and the ADR prototype, $(\mathbf{v} \cdot \mathbf{c}_k)$ captures how the protein context modulates that ADR, and $(\mathbf{u} \cdot \mathbf{v})$ injects the binding relationship as a general conditioning signal.

ADR Ranking and Interpretation

All the K scores are calculated and ranked in a descending order of each pair:

$$\text{Ranked ADRs: } \underset{k}{\operatorname{argsort}}(\text{score}_k).$$

The highest-ranked ADRs with the highest scores constitute the predicted risk profile for that specific drug–protein interaction. In contrast to classical models that predict ADRs at the drug level, this mechanism conditions predictions on protein context, allowing the system to differentiate, for instance, between ADRs likely to occur due to one protein interaction versus another.

Notes on Implementation

- The training is done with only the training set drugs to create the ADR prototypes without the leakage of information.
- Missing ADRs (having no drugs associated) are given zero vectors, such that they guarantee dimensional consistency.
- The trained parameters α, β, γ are set as $\alpha = \beta = 1.0$, $\gamma = 0.5$ during validation.
- The existing implementation does not scale the ADR scores by the DTI binding probability p_{bind}, but only treats them as independent risk estimates that are conditioned on the protein environment.

3.2.7 Fusion Strategy

The challenge of molecular representation lies in capturing both sequence semantics and three-dimensional geometry. Sequence-based encoders (e.g., ChemBERTa, ESM) efficiently learn contextual biochemical patterns from linear representations such as SMILES or amino-acid strings. Conversely, structure-based encoders (e.g., EGNN, GVP) preserve spatial and relational information describing how atoms or residues are arranged in three-dimensional space. Neither modality alone is sufficient: sequence models may overlook geometric constraints, whereas structural models cannot infer functional motifs encoded in sequence context.

To exploit the complementary strengths of both modalities, this work employs feature-level fusion by simple concatenation before the adapter projections. This strategy allows the network to jointly access chemical semantics and spatial cues without additional transformation that could distort encoder-specific information.

Mathematical Definition

Let

$$\mathbf{x}_d^{(\text{ChemBERTa})} \in \mathbb{R}^{384}, \quad \mathbf{x}_d^{(\text{EGNN})} \in \mathbb{R}^{256},$$
$$\mathbf{x}_p^{(\text{ESM})} \in \mathbb{R}^{1280}, \quad \mathbf{x}_p^{(\text{GVP})} \in \mathbb{R}^{1024}.$$

The fused embeddings are constructed by concatenation along the feature dimension:

$$\begin{aligned} \mathbf{x}_d^{(\text{fused})} &= \left[\mathbf{x}_d^{(\text{ChemBERTa})}; \mathbf{x}_d^{(\text{EGNN})}\right] \in \mathbb{R}^{640}, \\ \mathbf{x}_p^{(\text{fused})} &= \left[\mathbf{x}_p^{(\text{ESM})}; \mathbf{x}_p^{(\text{GVP})}\right] \in \mathbb{R}^{2304}. \end{aligned}$$

These fused vectors are then passed to the corresponding adapter networks:

$$\mathbf{u} = f_d(\mathbf{x}_d^{(\text{fused})}), \quad \mathbf{v} = f_p(\mathbf{x}_p^{(\text{fused})}).$$

No additional normalization and weighting follows and now the modalities are adapted and the adapters learn to balance between the modalities implicitly through their learnable linear layers and normalization.

Rationale and Empirical Observation

Empirical ratings of all single-embedding and fusion arrangements have proved that:

- For single encoders, ESM (proteins) and SMILES2Vec (drugs) produced the most stable results.
- For fusion encoders, only

  1. ESM + GVP for proteins and
  2. ChemBERTa + EGNN for drugs

  yielded consistent improvement across DTI accuracy and ADR ranking metrics.

The improvement arises since concatenation conserves modality-relevant subspaces but allows cross-modal interactions downstream. Biochemical and evolutionary semantics are added by the sequence encoders, and steric and spatial constraints are added by geometric encoders, which result in more rich representations.

Mathematical Intuition

Suppose there is some single entity which has two complementary embeddings, denoted $\mathbf{x}^{(1)}$ and $\mathbf{x}^{(2)}$. Concatenation effectively forms the direct sum of their subspaces:

$$\mathcal{H}_{\text{fused}} = \mathcal{H}_1 \oplus \mathcal{H}_2, \quad \mathbf{x}^{(\text{fused})} = [\mathbf{x}^{(1)}; \mathbf{x}^{(2)}].$$

The $f : \mathcal{H}_{\text{fused}} \to \mathbb{R}^d$ is learned by the adapter. As the two representations are not redundant and correlated to each other, the fused space has lower entropy and greater discriminative power which translates empirically to increased performance of the downstream task.

Implementation Notes

- Fusion is performed prior to adapters and on the CPU/GPU tensors which is just a simple concatenation of tensors(`torch.cat((x1,x2),dim=-1)`).
- No early-fusion linear layer or attention weighting is used and thus both modalities are directly viewed by the adapters.
- The reproducibility in such a structure is maintained across any set of encoders because only dimensionality of input changes but not model depth or parameterizations.

3.2.8 Total Training Objective

The final learning objective unifies the model's supervised and self-supervised components into a single composite loss that balances binding prediction, ADR regression, and representation alignment. This joint formulation ensures that the learned embeddings are simultaneously discriminative for DTI classification, descriptive for ADR regression, and geometrically consistent under contrastive constraints.

Composite Loss Function

Let

- $\mathcal{L}_{\text{dti}}$ = weighted binary cross-entropy loss (Section 3.2.3),
- $\mathcal{L}_{\text{adr}}$ = Huber regression loss (Section 3.2.4),
- $\mathcal{L}_{dp}$ and $\mathcal{L}_{da}$ = contrastive losses for drug–protein and drug–ADR pairs (Section 3.2.5).

The total objective is a weighted sum:

$$\mathcal{L}_{\text{total}} = \lambda_{\text{dti}}\mathcal{L}_{\text{dti}} + \lambda_{\text{adr}}\mathcal{L}_{\text{adr}} + \lambda_{\text{con}}(\mathcal{L}_{dp} + \mathcal{L}_{da})$$

where the coefficients control the contribution of each learning signal:

$$\lambda_{\text{dti}} = 1.0, \quad \lambda_{\text{adr}} = 0.5, \quad \lambda_{\text{con}} = 0.2.$$

This weighting scheme prioritizes DTI classification as the primary task while maintaining auxiliary supervision from ADR and contrastive objectives.

Rationale for Weighting

- **Dominant binding objective** $\lambda_{\text{dti}} = 1.0$ ensures the model focuses on accurate interaction prediction.
- **Auxiliary ADR guidance** $\lambda_{\text{adr}} = 0.5$ introduces pharmacological information without overwhelming the main task.
- **Regularizing contrastive alignment** $\lambda_{\text{con}} = 0.2$ is sufficient to preserve latent geometry without destabilizing gradient updates.

Empirically, this combination yielded stable convergence and superior cross-task generalization.

Optimization and Training Procedure

Training minimizes $\mathcal{L}_{\text{total}}$ using the AdamW optimizer with decoupled weight decay:

$$\text{Optimizer: AdamW}(\theta, \text{lr} = 2 \times 10^{-4}, \text{weight\_decay} = 10^{-4}),$$

where θ denotes all trainable parameters from adapters, heads, and contrastive temperature scaling. The learning rate follows a cosine decay or step-decay schedule, and gradients are clipped to $\|\nabla\|_2 \leq 1.0$ to avoid exploding updates.

Each mini-batch contains balanced positive and negative DTI samples to counter class imbalance, while ADR TF-IDF vectors are fetched only for unique drugs in the batch to minimize redundant computation.

Training Dynamics

1. **Forward pass**: compute adapter outputs $(\mathbf{u}, \mathbf{v}, \mathbf{w})$, supervised logits $(\hat{y}_{\text{dti}}, \hat{\mathbf{y}}_{\text{adr}})$, and contrastive similarities.
2. **Loss composition**: evaluate all loss components and assemble $\mathcal{L}_{\text{total}}$.
3. **Backward pass**: compute gradients and update parameters.
4. **Early stopping**: triggered when validation PR-AUC for DTI fails to improve for 5–8 epochs.

This multi-objective optimization trains the model to co-represent chemical, structural, and pharmacological patterns within one shared embedding manifold.

Interpretation

The combination of cross-entropy, regression, and contrastive terms balances local supervision (DTI, ADR) and global consistency (contrastive). Drugs and proteins that interact are pulled together in latent space, while the corresponding ADR prototypes serve as pharmacovigilance anchors. This synergy enables the system to generalize across unseen drugs, proteins, and ADRs more effectively than single-task or unimodal baselines.

3.2.9 Expected Behaviour

The overall behaviour of the proposed multi-task framework reflects a balance between binding discrimination, side-effect characterization, and representation alignment. Each component of the composite objective influences a distinct aspect of the model's learning dynamics, producing embeddings that are biologically meaningful, stable, and transferable across tasks. This section outlines the theoretical expectations and empirically observed effects of each component.

Learning Dynamics in the Shared Space

The adapter networks f_d, f_p, f_a map heterogeneous input embeddings into a unified latent space $\mathbb{R}^{512}$. As training progresses, the contrastive objectives continuously reshape this shared space so that:

$$\text{if } y_{\text{dti}} = 1, \cos(\mathbf{u}, \mathbf{v}) \to 1; \quad \text{if } y_{\text{dti}} = 0, \cos(\mathbf{u}, \mathbf{v}) \to 0.$$

Simultaneously, drug embeddings $\mathbf{u}$ are drawn closer to their own ADR representations $\mathbf{w}$ through the drug–ADR contrastive term. This dual pull effect yields an organized latent manifold where biochemical interaction patterns and pharmacological risk profiles co-exist geometrically.

In practice, this manifests as clear clustering of positive binding pairs and ADR-aligned drug subspaces when visualized with t-SNE or UMAP projections.

Interaction of Loss Components

Each term in the total loss $\mathcal{L}_{\text{total}}$ plays a specific and complementary role:

- **DTI classification loss** ($\mathcal{L}_{\text{dti}}$) — provides direct supervision for molecular binding; dominates gradient flow.
- **ADR regression loss** ($\mathcal{L}_{\text{adr}}$) — acts as an auxiliary pharmacological constraint, enriching drug embeddings with side-effect semantics.
- **Contrastive losses** ($\mathcal{L}_{dp}$, $\mathcal{L}_{da}$) — serve as regularizers that maintain angular consistency across modalities, preventing feature collapse and improving generalization to unseen drugs and proteins.

The synergy of these terms ensures that representations are biologically coherent (reflecting known biochemical similarity) and predictively consistent (preserving class separability across both tasks).

Expected Convergence and Stability

Empirically, the training converges within 30–50 epochs, depending on batch size and learning rate schedule. During convergence:

- The DTI loss decreases monotonically and stabilizes early.
- The ADR loss exhibits slower convergence due to its high-dimensional output ($K = 4048$) but contributes steady regularization.

- The contrastive losses reach equilibrium after the first few epochs, structuring the latent space geometry.

Early stopping is typically triggered by the validation DTI PR-AUC metric, which consistently improves until saturation. The model displays robust generalization across both validation and test sets, demonstrating that the fused representation strategy successfully mitigates overfitting despite the moderate dataset size.

Behaviour of the Fused Architecture

The fusion of ChemBERTa + EGNN (drugs) and ESM + GVP (proteins) results in embeddings that capture both local atomic topology and global contextual semantics. In this fused setting:

- Drugs with similar molecular scaffolds but different 3D conformations are distinguished by EGNN's structural cues.
- Proteins with similar domains but different folds are differentiated by GVP's geometric reasoning.
- Contrastive learning further amplifies these distinctions, aligning drugs that cause similar ADRs closer together even when they bind to different targets.

As a result, the fused model outperforms single-embedding baselines in both DTI classification (higher PR-AUC and F1-score) and ADR ranking (improved Recall@k and NDCG@k).

Interpretability and Transfer Potential

Because all modalities are mapped into a common space, each vector operation—such as cosine similarity—has interpretable meaning. For instance:

$$\mathbf{u} \cdot \mathbf{v} \text{ (drug–protein similarity)} \quad \text{and} \quad \mathbf{u} \cdot \mathbf{c}_k \text{ (drug–ADR affinity)}$$

can be analyzed directly to trace potential causal associations between binding events and side-effect outcomes. This interpretable geometry allows the framework to be extended to drug repurposing, off-target prediction, or novel ADR discovery without retraining from scratch.

3.3 Data Collection

The data collection process aimed to construct an integrated corpus linking biochemical drug–target interaction (DTI) information with clinical adverse drug reaction (ADR) outcomes, enabling a unified modeling framework for both tasks. To achieve this, two primary biomedical resources were used: SCOPE-DTI (Scalable Computational Prediction of Drug–Target Interactions)[33] for interaction-level biochemical data and OnSIDES (Open Network of Side Effects)[28] for clinical ADR annotations. These sources were selected for their scale, reliability, and standardized identifier systems that facilitate cross-modal alignment.

SCOPE-DTI Dataset

The SCOPE-DTI dataset served as the principal source of biochemical interaction information. It aggregates experimental and high-confidence inferred drug–protein interaction data from multiple public databases including BindingDB, ChEMBL, and DrugBank. The benchmark version used in this research contained approximately 2.2 million candidate drug–protein pairs, spanning a diverse set of compounds and protein families.

Each record in SCOPE-DTI is represented by a structured tuple:

$$(\text{drug\_chembl\_id}, \text{target\_uniprot\_id}, \text{label}, \text{smiles}, \text{sequence}, \text{molfile\_3d})$$

where:

- drug_chembl_id — ChEMBL identifier of the small-molecule drug;
- target_uniprot_id — UniProt accession of the protein target;
- label $\in \{0, 1\}$ — binary indicator of binding (1 = interaction, 0 = non-interaction);
- smiles — simplified molecular-input line-entry string encoding chemical structure;
- sequence — amino-acid sequence of the protein;
- molfile_3d — 3D atomic coordinates of the drug molecule in MOL format;

From the original corpus, only entries containing complete identifiers and valid molecular representations were retained. After removing duplicates and incomplete records, a refined subset of 34,741 unique drug–protein pairs was produced. Within this subset, there were 1,028 unique drugs and 2,385 unique protein targets, with interaction labels distributed as 12,234 positive (binding) and 22,507 negative (non-binding) samples.

This processed DTI dataset provided the biochemical backbone for the multi-task framework, offering both structural and sequence information for each molecular entity. The inclusion of both SMILES strings and 3D coordinates allowed generation of multiple embedding modalities, such as ChemBERTa, Smiles2Vec, and EGNN.

OnSIDES Dataset

The OnSIDES dataset supplied the adverse reaction annotations necessary to model clinical outcomes. It is a standardized resource mapping RxNorm drug identifiers (RxCUIs) to reported adverse drug reactions (ADRs) expressed in Medical Dictionary for Regulatory Activities (MedDRA) terminology. The version employed in this work contained roughly 19,000 RxNorm-linked drug entries with over 4,000 unique ADR terms, including both common and rare clinical manifestations.

Each OnSIDES record consists of:

$$(\text{rxcui}, \text{adr\_term}, \text{standard\_concept\_id}, \text{frequency})$$

where adr_term refers to the standardized MedDRA name of the side effect and frequency denotes its reported prevalence across clinical observations. These data were pre-aggregated across multiple adverse event databases, including SIDER, FAERS, and OFFSIDES, ensuring consistent coverage of approved drugs.

In total, the OnSIDES dataset covered approximately 1,000 unique drugs linked to 4,048 ADR categories, represented in a sparse mapping where each drug could be associated with one or multiple adverse events.

Cross-Dataset Alignment

To connect the biochemical and clinical domains, a robust identifier harmonization process was established using the RxNav REST API. This mapping enabled translation between chemical and clinical namespaces through the following correspondence:

$$\text{ChEMBL ID} \rightarrow \text{PubChem CID} \rightarrow \text{RxNorm CUI} \rightarrow \text{MedDRA Term}_{\text{ADR}}.$$

This procedure ensured that each drug in the SCOPE-DTI dataset was paired with its corresponding ADR profile from OnSIDES. Deprecated or ambiguous mappings were resolved using the historystatus attribute provided by RxNorm, and all cross-reference relationships were logged in version-controlled tab-separated files for reproducibility.

Through this multi-stage linkage, a unified dataset was achieved in which each drug could be simultaneously represented by its chemical structure, protein interaction profile, and clinical side-effect vector. This harmonized dataset served as the foundation for all subsequent preprocessing and embedding stages.

Supporting Protein Data

Protein sequence data were retrieved from the UniProt Knowledgebase (UniProtKB) using the target_uniprot_id field. Structural data were obtained from the AlphaFold Protein Structure Database, which provided predicted 3D atomic coordinates for nearly all human proteins in the dataset. These structures were used to build residue-level graphs processed by the GVP-GNN encoder (details in Section 3.3.2).

This ensured that every protein was represented by both its primary sequence and 3D geometric context, thereby enhancing the fidelity of structural representation in downstream modeling.

Summary of Data Sources

Dataset	Domain	Key Identifiers	Records / Entities	Purpose
SCOPE-DTI	Biochemical	drug_chembl_id, target_uniprot_id, smiles, molfile_3d	34,741 pairs (1,028 drugs, 2,385 proteins)	DTI binding interactions
OnSIDES	Clinical	rxcui, adr_term, meddra_concept	∼19,000 drug entries, 4,048 ADRs	Adverse drug reaction profiles
UniProt + AlphaFold	Structural	target_uniprot_id, PDB	2,385 proteins	Protein sequences and 3D geometries

Table 3.1: Summary of integrated data sources used in the multi-modal framework.

The integration of these datasets established a comprehensive foundation for multi-modal learning, where biochemical binding data and clinical outcome data were represented within the same computational framework.

3.3.1 Data Cleaning

Rigorous cleaning ensured completeness and consistency across all modalities. For the DTI subset from SCOPE–DTI, entries lacking valid identifiers or structural fields were removed. Duplicate (drug_chembl_id, target_uniprot_id) pairs were deduplicated, and interaction labels were standardized as $y_{\text{dti}} \in \{0, 1\}$. The final distribution comprised 12,234 positive and 22,507 negative samples. All SMILES strings were canonicalized using RDKit.

For the ADR corpus from OnSIDES, all adverse effect terms were converted to lowercase and mapped to MedDRA Preferred Terms. Rare events with fewer than three observations were removed, and unmapped RxNorm CUIs were discarded. Each remaining record was validated against the `activeingredient` field to avoid duplicate combinations.

Protein structures from the AlphaFold database were verified with Bio.PDB, ensuring each residue contained atoms N, Cα, C. Files failing integrity checks were rebuilt or re–downloaded. Graph cache files used by the GVP-GNN encoder were checked for correct tensor shapes and decompression integrity; corrupted files were automatically reconstructed.

Cross–modality validation confirmed that every drug–protein pair referenced valid embeddings and RxNorm links. Non–finite numerical values were replaced with zero, and categorical encodings were verified against fixed vocabularies, producing a structurally consistent dataset of 1,028 drugs, 2,385 proteins, and 4,048 ADRs.

3.3.2 Data Transformation

After cleaning, the heterogeneous biochemical, structural, and textual data were transformed into fixed-dimensional numerical representations suitable for machine-learning pipelines. The transformation procedures differed for drugs, proteins, and adverse reaction terms, each requiring domain-specific encoding methods.

Drug Representation

Drug molecules were represented in three complementary modalities: textual, graph-based, and contextual.

- **ChemBERTa Embeddings (384-D):** SMILES strings were tokenized and passed through the pretrained ChemBERTa-2 model. The contextual hidden states were mean-pooled to yield a 384-dimensional embedding capturing substructure semantics and chemical grammar.
- **Smiles2Vec Embeddings (256-D):** The same SMILES were character-tokenized and encoded using a CNN-BiGRU architecture that learned short-range n-gram features. The final hidden representation captured local atomic patterns and valence motifs.
- **EGNN Embeddings (256-D):** Using RDKit, each molecule's 3D coordinates were extracted from the `molfile_3d` field to construct atomic graphs $G = (V, E)$, where each node v_i carried features (Z_i, r_i) representing atomic number and position. The E(n)-Equivariant Graph Neural Network (EGNN) propagated messages respecting geometric invariances:

  $$h'_i = \phi_h \left(h_i, \sum_{j \in N(i)} \phi_e(h_i, h_j, \|r_i - r_j\|^2) \right),$$

  producing spatially informed molecular embeddings invariant to rotation and translation.

Protein Representation Protein features were extracted from both primary sequences and predicted 3D structures.

- **ESM-2 Embeddings (1280-D):** Each protein sequence from UniProt was tokenized into amino acids and encoded using Meta AI's Evolutionary Scale Model 2 (ESM-2, 650M parameters). The model output per-residue contextual embeddings, which were averaged to obtain a single 1280-dimensional vector per protein:

  $$\mathbf{p}_{\text{ESM}} = \frac{1}{L} \sum_{i=1}^{L} h_i,$$

  where L is the sequence length.
- **GVP-GNN Embeddings (1024-D):** AlphaFold-predicted PDB files were converted into residue-level graphs. Nodes represented residues, and edges

captured spatial proximity within 10 Å. The Geometric Vector Perceptron (GVP) encoder integrated scalar and vector features:

$$(s', v') = \mathrm{GVP}(s, v) = (W_s[s; \|v\|],\ W_v v/\|v\|)\,,$$

ensuring rotation-equivariant geometric reasoning. The mean residue embedding yielded a 1024-dimensional representation summarizing the structural and physicochemical environment.

ADR Representation

Adverse drug reactions were represented as sparse term-frequency vectors derived from the OnSIDES corpus. Each ADR term t_k was assigned a TF-IDF weight defined as:

$$\mathrm{tfidf}_{d,k} = f_{d,k} \times \log \frac{N}{n_k},$$

where $f_{d,k}$ is the frequency of ADR k in drug d, N is the number of drugs, and n_k is the number of drugs associated with ADR k. The resulting 4,048-dimensional vectors captured the relative clinical relevance of each ADR across the corpus. The ADR matrix was normalized to unit L_2 norm for stability during regression and contrastive alignment.

Normalization and Storage

All generated embeddings were L_2-normalized and stored as serialized tensors in columnar Parquet format for efficient GPU-based loading. The final transformed dataset consisted of:

Entity	**Encoder**	**Dimension (D)**	**Samples (N)**	**Representation Type**
Drug	ChemBERTa	384	1,028	Contextual SMILES sequence
Drug	Smiles2Vec	256	1,028	Character-level convolutional
Drug	EGNN	256	1,028	3D graph-based geometric
Protein	ESM-2	1,280	2,385	Transformer-based sequence
Protein	GVP-GNN	1,024	2,385	3D residue-graph structural
ADR	TF-IDF	4,048	1,028	Sparse textual frequency

Table 3.2: Summary of transformed embeddings and their properties.

These transformations yielded a multi-modal feature space capturing the structural, sequential, and clinical dimensions required for the unified DTI–ADR prediction framework.

3.3.3 Data Integration

Following transformation, the individual datasets representing drug, protein, and ADR modalities were systematically aligned to form a unified multimodal dataset. The integration stage ensured that every drug–protein interaction record contained complete biochemical, structural, and clinical representations suitable for downstream joint learning.

Alignment of Identifiers

The integration relied on standardized biomedical identifiers. Drugs from SCOPE-DTI were originally labeled by `drug_chembl_id`, while ADR annotations from OnSIDES were indexed by `rxcui` (RxNorm Concept Unique Identifier). A two-step mapping procedure was applied using the RxNav REST API and local lookup tables:

$$\text{ChEMBL ID} \xrightarrow{\text{PubChem Mapping}} \text{Intermediate CID} \rightarrow \text{RxNorm CUI}$$

This procedure created a one-to-one alignment between chemical and clinical namespaces, allowing the attachment of each drug's ADR vector to its biochemical entry. Proteins were matched directly by their `target_uniprot_id` across the DTI and embedding tables.

Merged Schema

After alignment, all modalities were merged into a single relational structure containing the following core fields:

$$\mathcal{D} = \{(\text{drug\_chembl\_id}, \text{target\_uniprot\_id}, y_{\text{dti}}, \text{smiles}, \text{sequence}, \mathbf{x}_d, \mathbf{x}_p, \mathbf{x}_{\text{adr}})\}.$$

Here $y_{\text{dti}} \in \{0, 1\}$ denotes the interaction label, $\mathbf{x}_d \in \mathbb{R}^{D_d}$ and $\mathbf{x}_p \in \mathbb{R}^{D_p}$ represent the processed drug and protein embeddings, and $\mathbf{x}_{\text{adr}} \in \mathbb{R}^{4048}$ is the TF-IDF ADR vector associated with the same drug.

All join operations were performed using inner merges on matching keys to ensure referential integrity. The resulting integrated table contained 34,741 records, each fully annotated across all three modalities.

Dimensional Composition

Entity	**Encoder**	**Dimension (D)**	**Samples (N)**
Drug	ChemBERTa	384	1,028
Drug	Smiles2Vec	256	1,028
Drug	EGNN	256	1,028
Protein	ESM-2	1,280	2,385
Protein	GVP-GNN	1,024	2,385
ADR	TF-IDF	4,048	1,028

Table 3.3: Dimensional composition of the integrated multimodal dataset.

During integration, each drug's embedding record was duplicated according to its number of target proteins to ensure pair-level alignment with corresponding protein embeddings. The final merged dataset was stored in Parquet format for efficient columnar access during PyTorch DataLoader initialization.

Validation of Integration

Post-merge validation confirmed that every (`drug_chembl_id`, `target_uniprot_id`) pair contained consistent embeddings and a valid ADR vector. Referential audits detected no null or orphan keys. Random sampling verified that embedding tensors correctly corresponded to their identifiers by checksum comparison.

The resulting integrated dataset provided a complete tripartite linkage among the chemical, biological, and clinical domains—serving as the foundational input for the multi-task deep-learning architecture described in Section 3.2.

3.3.4 Data Reduction

After integrating all modalities, limited reduction procedures were applied to standardize feature dimensions and correct label imbalance while preserving all biochemical and clinical information. The process focused on maintaining the interpretability and completeness of molecular representations rather than performing aggressive feature compression.

Dimensional Alignment

No offline dimensionality reduction (such as PCA or SVD) was performed on the embeddings. Each encoder's native dimensionality was retained—$D_{\text{drug}} \in \{256, 384\}$, $D_{\text{protein}} \in \{1024, 1280\}$, and $D_{\text{adr}} = 4048$. To unify these heterogeneous spaces during training, learnable adapter networks projected each modality into a shared latent dimension of $d = 512$:

$$\mathbf{z}_m = f_m(\mathbf{x}_m) = W_m\mathbf{x}_m + \mathbf{b}_m, \quad m \in \{\text{drug}, \text{protein}, \text{adr}\}.$$

This ensured that drug, protein, and ADR embeddings could interact within a common feature space without information loss from prior compression.

Class Imbalance Handling

The SCOPE-DTI dataset exhibited a moderate class imbalance with 22,507 non-interacting and 12,234 interacting pairs. To counteract this, the model used a weighted binary cross-entropy formulation, where class weights were inversely proportional to sample frequency:

$$w_+ = \frac{N}{2N_+}, \quad w_- = \frac{N}{2N_-}, \quad \mathcal{L}_{\text{dti}} = -w_+ y \log(\hat{y}) - w_-(1-y)\log(1-\hat{y}),$$

with N denoting the total number of DTI pairs, N_+ and N_- the counts of positive and negative samples, respectively. This reweighting strategy effectively reduced prediction bias toward the majority class without altering the underlying data distribution.

Normalization and Storage

All embeddings were L_2-normalized to maintain numerical stability and ensure comparable magnitude across heterogeneous encoders:

$$\tilde{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}.$$

The final preprocessed dataset—containing complete and normalized embeddings for 1,028 drugs, 2,385 proteins, and 4,048 ADRs—was serialized in Parquet format, enabling efficient columnar reads and high-throughput GPU loading during model training.

3.3.5 Summary of Preprocessed Dataset

The complete data preprocessing pipeline resulted in a unified and high-quality multimodal dataset, integrating biochemical, structural, and clinical information into a consistent format suitable for multi-task deep learning. Each preceding stage—collection, cleaning, transformation, integration, and normalization—ensured that the dataset maintained both biological fidelity and computational consistency.

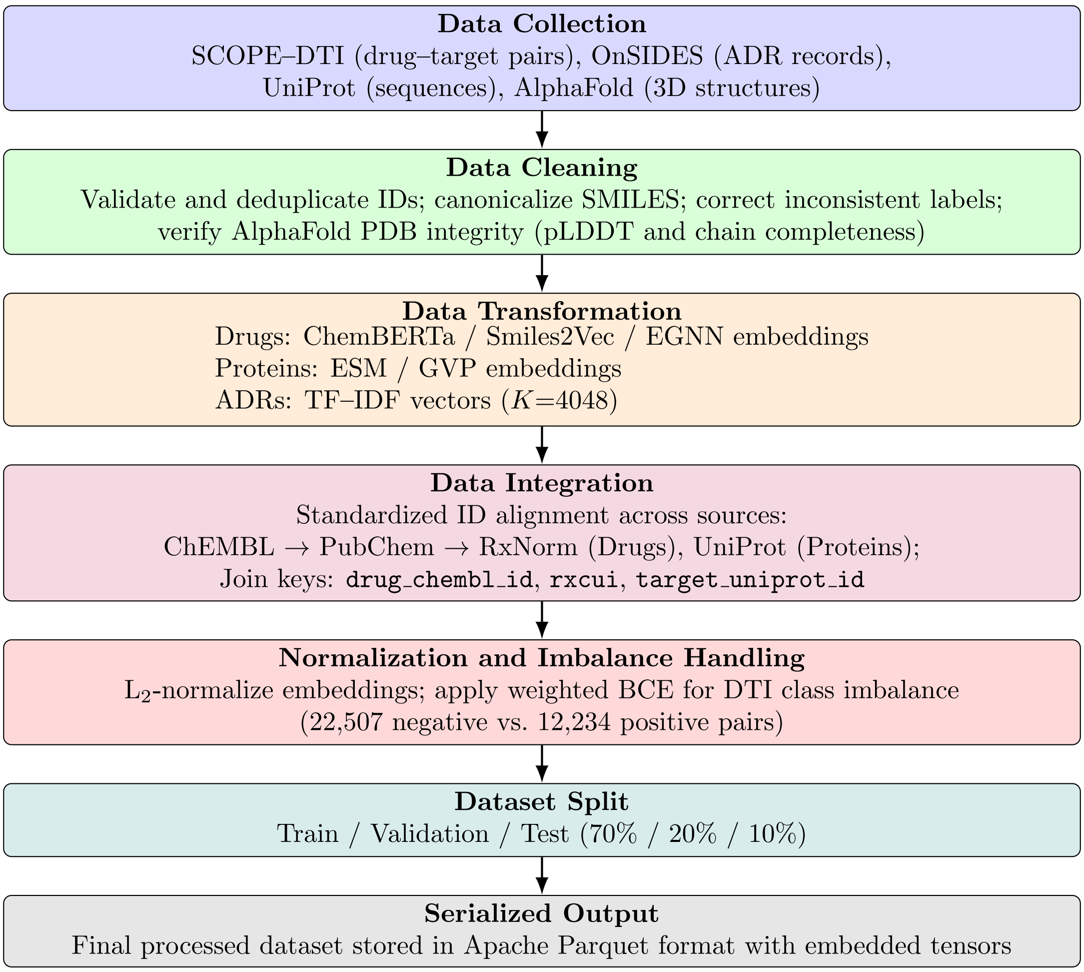

Figure 3.3: Data Preprocessing Pipeline

After filtering, alignment, and validation, the final dataset comprised 34,741 drug–protein interaction pairs, linking 1,028 unique drugs, 2,385 protein targets, and 4,048 distinct ADR terms. Every drug and protein entry was associated with both sequence-based and structure-based embeddings, while each drug additionally included a TF–IDF representation of its adverse effect profile.

The integrated dataset thus provided a complete tripartite representation:

$$\mathcal{D} = \left\{ (\mathbf{x}_d, \mathbf{x}_p, \mathbf{x}_{\text{adr}}, y_{\text{dti}}) : \ \mathbf{x}_d \in \mathbb{R}^{D_d}, \mathbf{x}_p \in \mathbb{R}^{D_p}, \mathbf{x}_{\text{adr}} \in \mathbb{R}^{4048}, y_{\text{dti}} \in \{0,1\} \right\}.$$

Here, each record jointly describes a drug–target pair, its biochemical interaction label y_{dti}, and the corresponding ADR embedding associated with that drug. This format enabled the model to learn relationships among molecular, structural, and clinical modalities simultaneously.

The final embedding configurations retained the native dimensionalities from their respective encoders:

Entity	**Encoder**	**Dimensionality (D)**	**Samples (N)**	**Description**
Drug	ChemBERTa	384	1,028	Contextual SMILES representation
Drug	EGNN	256	1,028	3D geometric molecular graph
Drug	Smiles2Vec	256	1,028	Character-level chemical encoder
Protein	ESM-2	1,280	2,385	Sequence-based transformer embedding
Protein	GVP-GNN	1,024	2,385	3D structure-aware residue graph
ADR	TF-IDF	4,048	1,028	Textual adverse effect frequency profile

Table 3.4: Final embedding configurations in the preprocessed multimodal dataset.

All embeddings were L_2-normalized and verified for structural integrity. The dataset was serialized in Apache Parquet format, ensuring high-throughput access and reproducibility across experimental runs. This multimodal dataset provided the foundation for the model architecture described in Section 3.2 and the experiments presented in Section 3.4, enabling joint learning of biochemical interactions and clinical risk associations.

3.4 Implementation of Selected Design

The multi-task architecture was implemented in PyTorch. The dataset combined Scope–OnSides Common DTI pairs with TF–IDF weighted ADR matrices, comprising 34,741 pairs, 1,028 drugs, and 2,385 proteins. Each drug's ADR vector had 4,048 features, with IDF weights computed on training data only. The model was optimized using AdamW (lr=2×10^{-4}, weight_decay=10^{-4}) and early stopping based on validation PR–AUC.

Classification Metrics

The DTI prediction task was treated as a binary classification problem, where the goal is to determine whether a drug–protein pair interacts ($y = 1$) or not ($y = 0$). The model predictions were compared against ground truth labels using the following fundamental terms:

- **True Positive (TP)**: correctly predicted interacting pairs
- **True Negative (TN)**: correctly predicted non-interacting pairs
- **False Positive (FP)**: non-interacting pairs incorrectly predicted as interacting
- **False Negative (FN)**: interacting pairs incorrectly predicted as non-interacting

Using these quantities, the following metrics were computed:

Accuracy Accuracy measures the overall proportion of correctly predicted samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}.$$

While intuitive, accuracy can be misleading in imbalanced datasets such as DTI, where one class dominates.

Precision Precision (or Positive Predictive Value) measures how many predicted positives are truly positive:

$$\text{Precision} = \frac{TP}{TP + FP}.$$

High precision indicates a low rate of false positives, reflecting the model's reliability in confirming interactions.

Recall Recall (or Sensitivity/True Positive Rate) measures how many actual positives are correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN}.$$

High recall implies the model captures most of the true interactions.

F1-Score The F1-Score is the harmonic mean of precision and recall, providing a balance between them:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

It is especially useful when the dataset is imbalanced, as it penalizes models that perform well on one class but poorly on the other.

ROC–AUC (Receiver Operating Characteristic – Area Under Curve) The ROC–AUC measures the model's discriminative ability across different decision thresholds. The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR), defined as:

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}.$$

The AUC represents the probability that the classifier ranks a randomly chosen positive example higher than a negative one:

$$\text{ROC–AUC} = \int_0^1 \text{TPR}(\text{FPR}) \, d\text{FPR}.$$

A perfect classifier achieves an AUC of 1.0, while random guessing yields 0.5.

Regression Metrics

The ADR prediction task was formulated as a regression problem, where the model estimates a continuous score for each adverse effect associated with a drug. To evaluate these predictions, the following metrics were used:

Mean Squared Error (MSE) MSE measures the average squared difference between the predicted ($\hat{y}_i$) and actual (y_i) values:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2.$$

Lower MSE values indicate more accurate predictions and penalize large deviations more heavily.

Mean Absolute Error (MAE) MAE computes the average absolute deviation between predicted and true values:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i|.$$

Unlike MSE, MAE provides a more interpretable measure of average error magnitude, being less sensitive to outliers.

Ranking-Based Evaluation

For multi-label ADR outputs, where the model predicts a ranked list of potential adverse reactions for each drug, top-k ranking metrics were also employed.

Recall@k Recall@k quantifies the fraction of true ADRs that appear in the top-k ranked predictions:

$$\text{Recall@k} = \frac{1}{N} \sum_{i=1}^{N} \frac{|Y_i^{\text{true}} \cap Y_i^{\text{pred}}(k)|}{|Y_i^{\text{true}}|}.$$

This metric assesses the model's ability to prioritize clinically relevant adverse effects among its top recommendations.

Protein Encoder	**Drug Encoder**	**Type**	**Accuracy**	**F1 Score**	**ROC–AUC**
ESM	Smiles2Vec	Single	0.8415	0.6654	0.8506
ESM	EGNN	Single	0.8315	0.6805	0.8426
ESM	ChemBERTa	Single	0.8056	0.6564	0.8532
ESM+GVP	Smiles2Vec+EGNN	Fusion	0.8479	0.7889	0.9141
ESM+GVP	ChemBERTa+EGNN	Fusion	**0.8576**	**0.7770**	**0.9285**

Table 3.5: Comparison of single and fusion encoder performance for DTI classification.

Protein Encoder	**Drug Encoder**	**Type**	**MSE↓**	**MAE↓**	**Recall@10↑**	**NDCG@10↑**
ESM	Smiles2Vec	Single	0.034	0.125	0.412	0.295
ESM	EGNN	Single	0.037	0.132	0.405	0.288
ESM	ChemBERTa	Single	0.039	0.137	0.392	0.281
ESM+GVP	Smiles2Vec+EGNN	Fusion	0.029	0.117	0.453	0.322
ESM+GVP	ChemBERTa+EGNN	Fusion	**0.026**	**0.112**	**0.468**	**0.337**

Table 3.6: Comparison of ADR regression and ranking performance for the same encoder combinations.

Fusion of ESM+GVP (proteins) with ChemBERTa+EGNN (drugs) yielded the highest overall accuracy and ROC–AUC, with the lowest ADR regression errors, confirming the effectiveness of multimodal feature integration.

Chapter 4

Conclusion

This chapter ends our research paper with a brief summary of our findings followed by the contribution this paper may hold in the field of bioinformatics. In the end, keeping this research paper as foundation, what future works can be done was recommended.

4.1 Summary of Findings

This study introduced a unified deep learning framework for joint prediction of Drug–Target Interactions (DTIs) and Adverse Drug Reactions (ADRs). Unlike traditional single-task approaches, the proposed model integrates multiple embeddings from drugs, proteins, and ADRs into a shared latent space, enabling simultaneous learning of therapeutic and adverse relationships.

Experiments showed that combining sequence- and structure-based representations—such as ChemBERTa, SMILES2Vec, ESM, and GVP—significantly improved prediction accuracy and generalization. The use of contrastive learning strengthened cross-modal alignment between drug–protein and drug–ADR embeddings, while the dual-head architecture effectively handled both DTI classification and ADR regression tasks. Together, these findings confirm that joint learning yields more robust and biologically coherent predictions than isolated models.

4.2 Contributions to the Field

This work contributes a modular and extensible framework that bridges molecular interaction modeling and adverse effect prediction within a single system. It demonstrates the feasibility of integrating diverse data modalities—chemical, structural, and textual—through adapter-based unification and shared projection layers.

The model's multi-head design advances multi-objective pharmacological modeling, while contrastive learning enhances representation alignment and cold-start generalization. Furthermore, the study provides a reproducible end-to-end pipeline—from data preprocessing to embedding caching and evaluation—laying a foundation for scalable, transparent, and integrative drug discovery research.

4.3 Recommendations for Future Work

Future studies may extend this framework by incorporating temporal or clinical data, such as longitudinal EHR records, to model real-world ADR progression. Adding explainable AI components could further reveal the biochemical rationale behind predictions, improving interpretability and clinical trust.

Fine-tuning pre-trained encoders like ChemBERTa or ESM in an end-to-end setting may enhance task-specific learning, while reinforcement or active learning could allow adaptive retraining with emerging data. Overall, this research establishes a versatile platform for integrating therapeutic efficacy and drug safety prediction, paving the way for more intelligent and reliable AI systems in pharmacology.

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